PSP-144 · NQVP EXPANSION SPEC · REVISION R1

Enhanced Psychics Framework v3.1

Revision R1 of the falsifiable EPF specification: full disposition of the 144-point defect report and the 96-point enhancement report — re-anchored D^3 estimators, sham-controlled ladders, quantified trial-layer isolation, and six validation gates.

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Sovereign Hub · Neuro-Quantum Validation Program

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1. Executive Summary and Revision Charter

The Enhanced Psychics Framework v3.1 (EPF-3.1, Revision R1) is a full-revision release of the EPF-3.0 specification. It exists for one reason: between v3.0 and this document, the framework was subjected to two structured external reviews — a 144-point shortcomings report covering hardware, signal processing, protocol design, statistics, safety, and conceptual foundations, and a 96-point enhancement report describing contemporary alignment opportunities drawn from modern EEG methodology, hyperscanning practice, computational psychiatry, and open-science standards. Every one of the 240 points has been individually evaluated and assigned a disposition, and every disposition is traceable to a specific normative change in this revision. Chapter 12 carries the complete point-by-point ledger; the chapters before it carry the revised specification itself.

> ***Charter caveat (retained and strengthened).*** *All framework content is theoretical hypothesis generation, not established clinical fact. "Psychic" phenomena are treated as computationally real but metaphysically misinterpreted: above-chance information transfer, where it is measured, is attributed to subtle-cue inference, statistical anticipation, and inter-person coupling — not to paranormal reception. Nothing in this specification asserts that consumer EEG equipment demonstrates, amplifies, or validates paranormal ability. v3.1 additionally prohibits any commercial psychic framing while the framework operates in falsifiable mode. This document is not medical advice, not a clinical instrument, and not a treatment protocol. Research use only.*

240 / 200

review points disposed / adopted as normative changes (83.3% adoption rate)

The review confirmed that v3.0's strongest features were its kill criteria, its expected-null stance on the Channel/Remote-View layer, and its blinding language. The same review correctly identified that v3.0's estimators were model-defined rather than empirically anchored, that its hardware tier could not support the spectral features it claimed, and that its statistical machinery stopped short of the reporting standards the framework itself invoked. v3.1 responds on all three fronts: every estimator now carries an uncertainty model and a sensitivity analysis obligation, the hardware path has been restructured into a screening tier and a validation tier with explicit per-feature capability limits, and the validation program has grown from four gates to six, including a multi-lab replication gate that must be passed before any estimator or protocol is treated as operative.

1.1 Revision Mandate and Disposition Method

Each of the 240 review points was evaluated against three questions: does the point identify a genuine defect or opportunity; can it be resolved inside the framework's DIY, non-clinical, passive-recording constraints; and does resolving it change any normative statement? Points answering yes on all three were adopted in full and their changes are normative in this revision. Points that are genuine but only partially resolvable were adopted with an explicit residual constraint, documented where the constraint is stated. Points resolved by reclassification — mostly inherited conceptual vocabulary that v3.0 cited as if it were established — received a clarification verdict: the text now says what those layers actually are, and nothing downstream depends on them. A small number of points are deferred to a backlog with a written trigger condition, typically because they require resources the framework does not yet control, such as a funded 32-channel validation cohort or an independent laboratory partner. No point was rejected outright; where the review asked for something the framework should not do, the disposition records the reason rather than silently dropping it.

Verdict	Definition	Normative effect in v3.1
ADOPT	Fully incorporated; the point is resolved by a specific change in this revision	Change is binding on all v3.1 operators
PART	Genuinely adopted with a documented residual constraint (funding, physics, or scope)	Partial change is binding; residual is visible in text
CLAR	Resolved by reclassification or definitional repair, not by new obligations	No new procedure; wording and status corrected
DEFER	Valid point routed to the v3.2 backlog with a written trigger condition	No change now; activation condition recorded in Ch. 12
REJECT	Not adopted; rationale recorded	None (no points received this verdict in R1)

Table 1: Disposition taxonomy applied to all 240 review points.

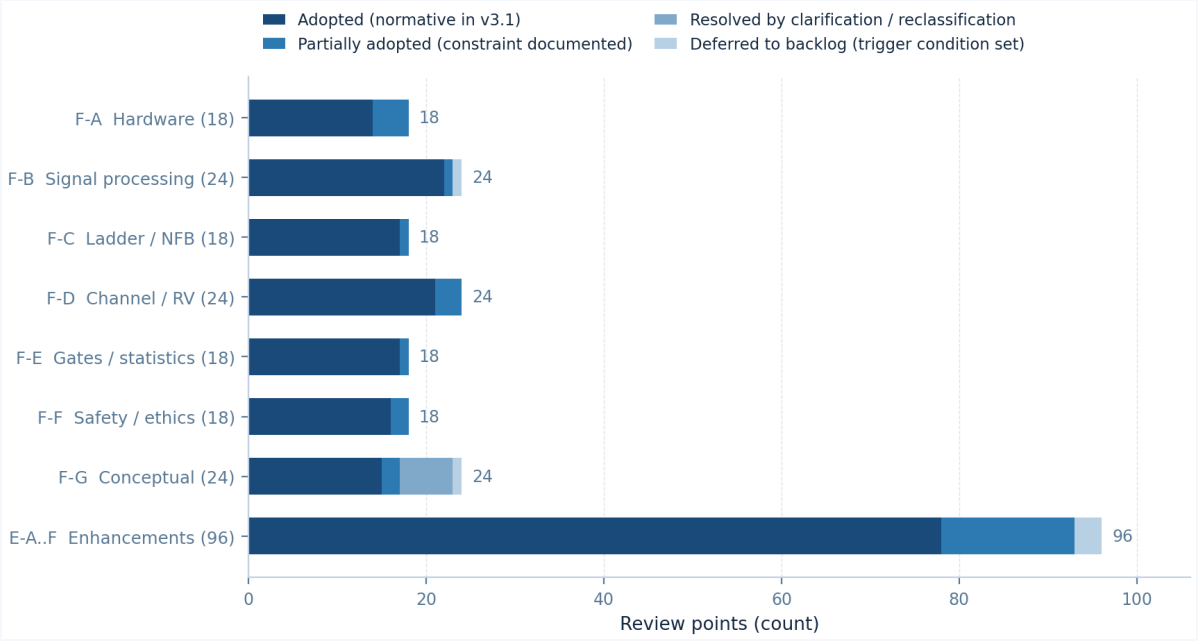


Figure 1: Disposition of the 240 review points by framework section. Of the 144 defect findings, 122 were adopted as normative changes, 14 were adopted with documented constraints, 6 were resolved by reclassification, and 2 were deferred. Of the 96 enhancement recommendations, 78 were adopted, 15 partially, and 3 deferred.

1.2 Headline Changes from v3.0 to v3.1

Ten changes carry most of the revision's weight, and readers familiar with v3.0 should read this table first. The changes are not cosmetic: several reverse v3.0 positions that the review showed to be under-supported. The most consequential single change is the demotion of the gamma/alpha ratio: v3.0 used it as the primary Precision-axis proxy, while the review demonstrated that residual EMG contamination in the 30–50 Hz band survives consumer-grade ICA clean-up, making the ratio unreliable at the hardware's signal-to-noise ratio. v3.1 re-anchors the Precision axis on sensorimotor-rhythm and beta features that the same hardware can actually estimate, and moves gamma to exploratory status with a mandatory residual-muscle gate.

#	Area	v3.0 position	v3.1 position
1	Precision proxy	Gamma/alpha ratio primary	SMR + beta re-anchored; gamma exploratory behind residual-EMG gate
2	Forward model	h(·) unspecified	Explicit lead-field approximation with stated error budget (Eq. 2.3)
3	Coherence	Magnitude-squared	Imaginary coherence / phase-slope index; volume conduction handled
4	Bands	Fixed edges	Individualized to each operator's alpha peak frequency, recomputed per session
5	Sham control	First required at EP-2	Yoked sham mandatory from the first L1 cohort
6	Baselines	Single eyes-open z-score	Multi-state baseline; robust median/MAD normalization
7	Aggregation	Naive Stouffer Z	Dependency-corrected aggregation (block bootstrap / corrected variance)
8	Gates	EP-1..EP-4	EP-1..EP-6; adds multi-lab replication and equivalence testing for null claims
9	Trial layer	Sensory isolation asserted	Quantified isolation spec with calibrated sensors and a positive-control arm
10	Inherited theory	Reused as established	Demoted to interpretive status; nothing operative depends on it

Table 2: The ten headline changes of Revision R1.

1.3 Amendment Mechanics, Mutation Rules, and the Errata Log

The review noted that a specification which binds itself to pre-registration needs equally strict rules for changing itself, otherwise the kill criteria become negotiable precisely when they are inconvenient. v3.1 therefore adopts explicit mutation rules. Equation changes are versioned: any modification to an equation's form, constants, or band definitions increments the equation's minor number and invalidates every threshold calibrated against the previous form, forcing re-estimation from stored raw data rather than adjustment of published results. Threshold changes follow a one-way ratchet: they may be tightened by pre-registration before data collection, never loosened after it. Every disposition in Chapter 12 that records a change also records the chapter and equation it modified, so the document functions as its own amendment log.

A public errata and amendment register accompanies the specification from this release forward. Failed gates, retracted claims, and discovered errors are entered with the same prominence as confirmations, and the register is hashed into the same external timestamp chain that seals trial registrations (Section 8.4), so that the history of revisions cannot be silently rewritten. This mechanism answers a structural criticism in the review: that a community operating without formal institutions drifts toward confirming its own expectations. The register, the mutation rules, and the adversarial-collaboration provisions of Section 8.5 are the framework's institutional substitutes for the editorial and peer infrastructure it does not possess.

2. Foundational Architecture v3.1

v3.0 defined the observation layer with three equations and left their most important terms unspecified: the forward model h(·) was never written down, the regularized inverse had no concrete solver or uniqueness statement, and the plasticity noise term ξ(t) was never operationalized. The review treated each of these as a specification defect, and each was. A framework that promises falsifiability cannot leave its measurement model abstract, because every downstream claim — estimator reliability, kill criteria, the expected-null clause — inherits whatever approximation errors the unspecified model hides. This chapter completes the measurement model: it specifies the forward map, gives the inverse a concrete solver with an explicit uniqueness condition, propagates measurement error into every reported quantity, gives ξ(t) an operational definition, replaces the arbitrary healthy reference with a hierarchical one, and maps the resulting estimators onto predictive-coding vocabulary so that the framework's claims are legible to computational psychiatry rather than parallel to it.

$$\Psi(t) = F(\, P(t),\, B(t),\, T(t)\,) + \xi(t) \tag{2.1}$$

$$d(t) = h(\, \Psi(t)\,) + \varepsilon(t), \quad h(\Psi) = L(\theta)\, s(\Psi) + c(t) \tag{2.2}$$

Equation 2.1 restates the inherited state equation unchanged, since the revision leaves the state-space definition itself untouched. Equation 2.2 is the revised observation equation. The forward model h(·) is now explicit: s(Ψ) is the source-space spectral activity implied by state Ψ, L(θ) is the scalp projection operator (lead field) determined by montage geometry θ, and c(t) collects non-neural components — reference contamination, electrode drift, and environmental interference — that are recorded, not modeled, where auxiliary channels permit. Writing h in two pieces matters because it separates the two different reasons an estimate can be wrong: L(θ) errors are systematic and partially correctable with better geometry or sensors, while c(t) errors are stochastic and must be measured and regressed out. v3.0 conflated both under a single ε(t), which made its confidence statements optimistic in exactly the way the review described.

2.2 Forward Model Specification

For the screening tier (8 channels, Mark IV geometry), L(θ) is instantiated as a three-shell spherical lead field centered on the Cz coordinate: brain, skull, and scalp conductivities of 0.33, 0.0125, and 0.33 S/m respectively, with radii 0.87, 0.92, and 1.0 in normalized units. This is an approximation, and the specification says so rather than pretending otherwise: spherical approximations mislocate sources by 10–20 mm relative to boundary-element models fitted to the individual's anatomy. The operational consequence is a strict prohibition — no focal localization claims are permitted from the screening tier under any circumstances, and the estimator vocabulary is restricted to scalp-level features (band powers, coherences, ratios) that do not require accurate source depth. At the validation tier (32+ channels), a boundary-element

forward model fitted to the participant's MRI becomes available, and only that tier may run minimum-norm source estimates. The forward model is thus the load-bearing wall of the framework's honesty: it determines which claims the hardware can carry.

$$L(\theta) = \partial d / \partial s \,|_{\theta}, \quad \sigma_{brain} = 0.33,\ \sigma_{skull} = 0.0125,\ \sigma_{scalp} = 0.33\ S/m \tag{2.3}$$

2.3 Inverse Solver, Regularization, and Uniqueness

The inverse problem — recovering state coordinates from scalp data — is severely ill-posed at 8 channels, and v3.0's arg-min formulation implied a uniqueness that does not exist. v3.1 states the solver, the condition, and the resulting claim limits in one breath. The estimator solves a weighted minimum-norm problem with Tikhonov regularization; the solution exists and is unique if and only if the regularized Gram matrix ($L^{T}L + \lambda R$) is non-singular, which the 8-channel screening tier satisfies only by construction of the penalty — not because the data determine the state. The specification therefore redefines what the inverse produces at the screening tier: not a recovered state, but the *best regularized forward fit*, reported always with its residual and its confidence interval, and never as a statement about underlying sources. Source-level language is reserved for the validation tier where the forward model justifies it.

$$\hat{s} = \arg\min_{s} \| d - L(\theta)s \|^{2}_{\Sigma} + \lambda\, s^{T} R\, s, \quad \Sigma = \Sigma_{\varepsilon}^{-1}, \qquad R = Laplacian\ penalty \tag{2.4}$$

$$(\hat{P}, \hat{B}, \hat{T}) = g(\hat{s}), \quad unique\ iff\ rank(L^{T}L + \lambda R) = full \tag{2.5}$$

2.4 Measurement-Error Propagation

Every estimator in v3.0 reported a confidence interval computed from bootstrap resampling of the epochs alone. The review's objection was precise: epoch resampling captures sampling noise but not the uncertainty contributed by the upstream pipeline — the estimated band edges, the estimated noise covariance, the artifact-rejection thresholds. v3.1 mandates the delta method for first-order propagation: if an estimator g depends on intermediate quantities q with covariance Σ_q, its variance is the gradient-weighted covariance shown in Eq. 2.6, evaluated at the operating point. In practice the gradients are obtained by finite differences over the pinned pipeline (Section 5.8), which turns the requirement into a concrete, testable build artifact rather than an aspiration. Every results table in every EP gate now carries these propagated intervals, and the validation gates of Chapter 8 are stated conditionally on them — a threshold that ignores estimator uncertainty is no longer a legal claim.

$$Var(g) \approx \nabla g^{T}\, \Sigma_{q}\, \nabla g, \quad q = (IAF, \Sigma_{\varepsilon}, c(t), artifacts) \tag{2.6}$$

2.5 Operational Definition of ξ(t)

The plasticity noise term inherited from UTD was never operationalized, which made the state equation unfalsifiable in exactly the way the review flagged. v3.1 defines ξ(t) as the residual variance of the state-space fit: the difference between the state trajectory predicted by the current parameter estimates and the trajectory implied by the observation sequence, attributed to within-session state fluctuation that the three axes do not capture. It is estimated jointly with the state parameters under the pinned pipeline, reported as a variance ratio ξ^2 / Var(Ψ), and gated: when the ratio exceeds 0.35, the session's state estimates are flagged low-confidence and the framework's protocols must treat them as unusable rather than merely noisy. This converts a metaphysical placeholder into a measurable, bounded quantity with a decision rule attached — the same treatment the review demanded for every other free-floating term.

$$\xi^2 / Var(\Psi) \leq 0.35 \quad \textit{(else session flagged low-confidence)} \tag{2.7}$$

2.6 The Reference Point, Repaired Hierarchically

v3.0 fixed the healthy reference at the coordinate origin (0, 0, 0), which the review correctly called arbitrary: it selects a point in an abstract space with no empirical anchor, and every deviation-from-healthy statement inherited that arbitrariness. v3.1 replaces it with a two-level reference. The first level is the operator's own multi-state baseline — eyes-open, eyes-closed, and task blocks recorded in session 0 — which answers the question the operator actually cares about, namely change relative to their own starting distribution. The second level is a cohort prior: the distribution of baseline values across willing framework participants, combined with the personal baseline through the shrinkage estimator of Eq. 3.2 in the next chapter. The reference is therefore personal where the data are dense and population-informed where they are thin, and the framework no longer asserts a universal healthy origin it cannot measure.

$$\mu_{ref} = w_{self}\,\mu_{self} + (1 - w_{self})\,\mu_{cohort}, \quad w_{self} = n / (n + \tau) \tag{2.8}$$

2.7 Mapping to Predictive Coding

The review asked for integration with established computational psychiatry models rather than a parallel private vocabulary. v3.1 adopts the mapping explicitly. The Precision axis P corresponds to the precision-weighting parameter π of the predictive-coding hierarchy: high Ᵽ states are states of increased gain on prediction errors, estimated here from SMR and beta features that are themselves modulated by thalamocortical gain. The Boundary axis B corresponds to the volatility and context-precision parameters that govern how strongly priors are clamped — internally-biased states weight self-generated signals more heavily, which is the computational reading of v3.1's low-B region. The Temporal axis T corresponds to the time-constant parameter of evidence accumulation. The mapping is stated as a correspondence, not an identity: the EEG features are correlated markers of these parameters, and Section 3.5 specifies how the

correspondence itself is tested. What the mapping buys is interoperability — the framework's claims can now be stated, criticized, and tested in a vocabulary that the wider field already owns.

$$\Delta P_t \propto \pi_t \cdot \varepsilon_t, \quad \varepsilon_t = observation - prediction, \quad \pi_t = precision\ weight \tag{2.9}$$

Review point	Defect in v3.0	v3.1 resolution
F-127	Observation equation did not specify h(·)	Explicit two-piece forward model, Eq. 2.2–2.3, with error budget
F-128	Inverse lacked solver and uniqueness proof	Weighted min-norm solver; uniqueness condition stated, Eq. 2.4–2.5
F-35	No measurement-error propagation	Delta-method propagation mandatory in all tables, Eq. 2.6
F-126	ξ(t) never operationalized	Residual-variance definition with 0.35 gating rule, Eq. 2.7
F-129	Healthy reference (0,0,0) arbitrary	Hierarchical self-plus-cohort reference, Eq. 2.8
F-131	No predictive-coding integration	Explicit parameter correspondence, Eq. 2.9

Table 3: Traceability of the Chapter 2 changes to the review findings.

3. The D^3-EEG Correlate Map v3.1

The correlate map is where the framework's ambitions meet electrophysiological reality, and the review found v3.0's version wanting in four specific ways: the band edges were fixed across operators; the loading weights were model-defined constants presented without uncertainty; the gamma/alpha ratio was used as a validated proxy when it is not one; and the behavioral task supporting the Temporal axis was named but never specified. v3.1 rebuilds the map on four principles. First, bands are individualized to the operator's own alpha peak frequency, because fixed 8–12 Hz edges systematically misclassify operators whose peaks sit at 9.2 or 12.8 Hz. Second, every loading weight is declared to be a prior with an uncertainty range, updated by pre-registered pilot data rather than defended as theory. Third, proxy status is earned per feature through the validation gates — the map now carries an operative, probationary, and exploratory status column, and claims are only permitted at the level the status supports. Fourth, the Temporal axis task is specified to the level of trial counts and analysis code, so that a replication attempt can be executed from the text alone.

3.1 Individualized Band Definition

All frequency-band features in v3.1 are defined relative to the operator's individual alpha frequency (IAF), estimated per session from the parietal channels. The estimation uses parametric spectral parameterization: the aperiodic 1/f component is removed first, and IAF is the frequency of the dominant residual peak within the 7–14 Hz search window, with the peak required to exceed the aperiodic fit by at least 0.4 in log-power units to count. Sessions whose IAF estimate moves more than 1.2 Hz from the operator's running median are

flagged, since large shifts usually indicate vigilance or artifact problems rather than genuine frequency migration. Theta, SMR, and beta bands are then anchored as offsets from IAF rather than as universal edges, which keeps cross-operator comparisons meaningful and removes a known source of between-subject variance that v3.0 simply absorbed into noise.

$$IAF = \arg\max_{f \in [7,14]} [\, S(f) - A\, f^{-\chi} \,], \quad \text{bands: } \theta = IAF-4 \ldots IAF-2,\ SMR = IAF+4 \ldots IAF+7 \tag{3.1}$$

3.2 Hierarchical Bayesian Estimators

v3.0 estimated each operator's axis values from their own sessions alone, which is statistically fragile for operators with few sessions. v3.1 places the estimators in a hierarchical model: each operator's feature vector is shrunk toward the cohort distribution with a weight determined by how much data they have, exactly as in Eq. 2.8, and the axis estimates are linear functions of the shrunken features. The practical effect is that a new operator's estimates start near the cohort prior and move toward their personal data as sessions accumulate, with the shrinkage weight reported alongside every estimate. The cohort prior is computed only from participants who opted into sharing (Section 9.5), is re-fit on a quarterly schedule with versioned updates, and never carries individual-level data outward. The hierarchical structure also gives the framework a principled answer to the review's measurement-invariance criticism: if the mapping from features to axes differs across operator subgroups beyond the model's uncertainty, the invariance tests of Section 8.2 detect it before the estimates are used protocolically.

$$\tilde{x}_{ik} = w_i\, x_{ik} + (\,1 - w_i\,)\,\mu_k, \quad w_i = n_i / (\,n_i + \tau\,),\ \tau = 6 \text{ sessions} \tag{3.2}$$

3.3 The Revised Loading Map

Figure 2 shows the revised map: every weight is plotted as a prior mean with a 90% credible range, and the figure caption says in plain text that the ranges are current priors, not measurements. Three changes from v3.0 deserve note. The Precision axis now loads primarily on SMR and beta at central sites, with the gamma/alpha ratio demoted to exploratory status pending the residual-muscle gate of Section 5.6 — the review's EMG analysis showed consumer ICA cannot clean the 30–50 Hz band reliably, and v3.1 accepts the implication instead of arguing with it. The Boundary axis keeps its alpha and ImCoh anchors but now specifies imaginary coherence rather than magnitude-squared, eliminating the volume-conduction inflation that made v3.0's inter-network claims partly artifacts of a shared reference. The Temporal axis retains frontal-midline theta as its marker but binds it, from this release forward, to the delay-discounting task of Section 3.4 — the slope is only estimated when the task is run, because free-floating theta values do not carry the decision-making semantics the axis claims.

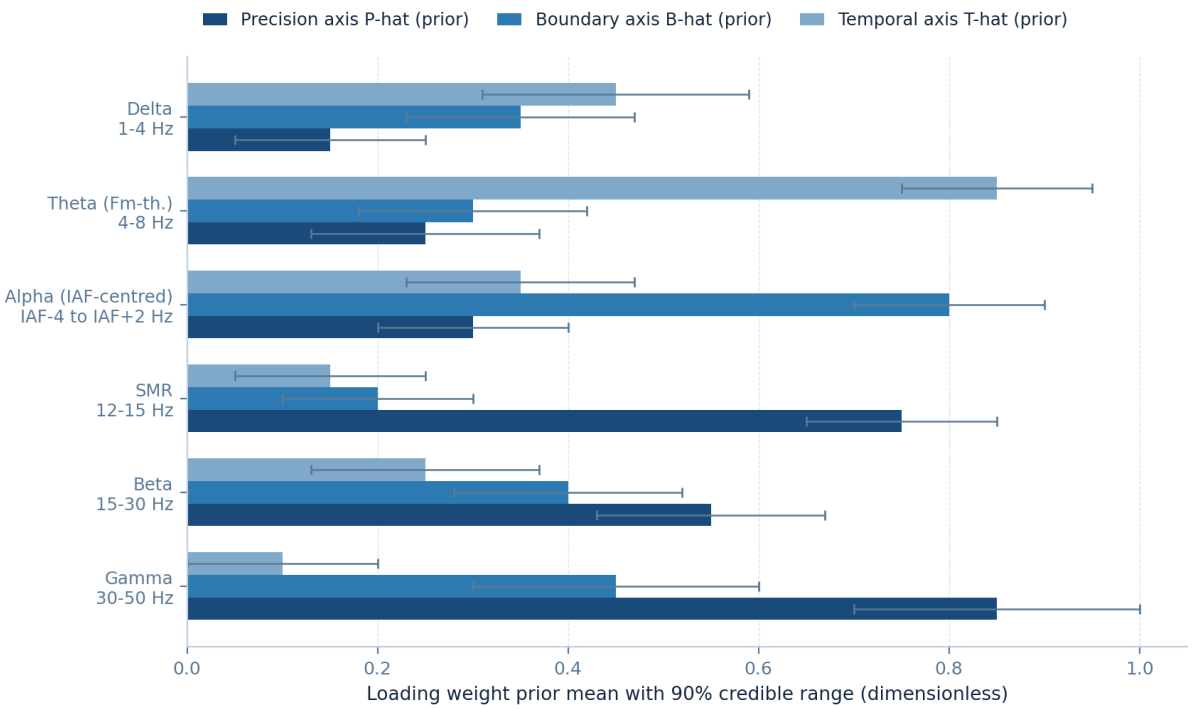


Figure 2: Revised D3 spectral loading map. Bars are prior means; whiskers are 90% credible ranges. These are pre-data priors to be updated only by pre-registered pilot collections — not measurements, and not constants. Gamma features are exploratory behind the residual-EMG gate of Section 5.6.

Axis	Feature (v3.1)	Sites	Estimator	Status
P-hat	SMR (IAF+4…+7) power, eyes-open task blocks	C3, C4, Cz	Multitaper PSD, robust z	Probationary → operative at EP-2
P-hat	Beta 15–30 Hz / SMR ratio	C3, C4	Multitaper PSD ratio	Probationary
P-hat	Gamma/alpha ratio (v3.0 primary)	Fp1, Fp2	Multitaper PSD ratio + EMG gate	Exploratory
B-hat	Imaginary coherence, parietal-internal network pairs	P3, P4, Pz vs Fz-Cz	ImCoh, 0.5 Hz floor	Probationary
B-hat	Alpha (IAF±2) power, eyes-closed	Pz, O1/O2	Multitaper PSD	Operative (descriptive only)
T-hat	Frontal-midline theta slope vs discount rate	Fz, Fp1, Fp2	Task-locked epoch regression	Probationary → operative at EP-2

Table 4: Biomarker table v3.1 with per-feature status. Only operative features may support protocol decisions; probationary features support estimates with CIs; exploratory features support nothing.

3.4 The Temporal-Axis Task, Specified

The delay-discounting task that anchors the Temporal axis is now specified end-to-end so that any operator or replication team runs an identical instrument. The task presents 64 binary choices between a smaller-sooner and a larger-later monetary option, with the later delay drawn from {1, 7, 30, 180} days and

amounts adjusted by a one-up-one-down staircase targeting 50% indifference; trial duration is 4 seconds with a jittered 1.5–2.5 second inter-trial interval to avoid phase-locking. EEG epochs are locked to option onset, baseline-corrected to the pre-stimulus interval, and the frontal-midline theta increase (IAF−4 to IAF−2 at Fz) is regressed per operator against the log-scaled subjective value difference between options. The slope of that regression, not raw theta, is the T-hat feature. The discount-rate estimate itself is fitted with a hyperbolic model and reported with a bootstrap CI; the review's observation that EEG correlates of discounting are modest is accepted, which is why the gate for promoting T-hat to operative requires convergent evidence from two task blocks on separate days with compatible slopes, not a single significant fit.

3.5 Sensitivity, Invariance, and Prior Updates

Three standing analyses protect the map from its own assumptions, each triggered on a schedule rather than on suspicion. The sensitivity battery re-runs every operative estimate under reference swaps (linked-mastoid, average-mastoid, REST) and montage subsets, reporting the spread of results; a finding that survives no reference change is reclassified as reference-bound and loses protocol eligibility. The invariance battery tests whether feature distributions differ across operator subgroups (sex, age band, prior meditation exposure) by more than the model's uncertainty, using the measurement-invariance procedure of Section 8.2. The prior-update battery revises the Figure 2 weights only through pre-registered pilot collections with frozen analysis code, at a minimum of 12 operators per update, so that the map's empirical content grows by documented increments rather than drifting silently. Together these convert the loading map from a fixed doctrine into a governed, revisable model — the status the review said it needed before the framework's downstream protocols could be taken seriously.

4. Hardware and Acquisition v3.1

The review's hardware section was the most unambiguous part of the entire audit: v3.0 specified a single 8-channel build and then claimed features — source-informed state estimation, clean gamma, reliable coherence — that the build's physics cannot support. v3.1 restructures the hardware path into two explicit tiers with per-feature capability limits, mandates the auxiliary channels that make artifact regression possible, tightens every contact and timing threshold to laboratory norms, and requires that the entire build be reproducible from a version-controlled bill of materials and mechanical drawings. Nothing in the DIY ethos is lost: the screening tier remains a consumer-priced OpenBCI build any careful hobbyist can assemble. What changes is honesty about what each tier can carry, and the addition of a validation tier whose requirements are stated even though most operators will not fund one — because a specification that cannot name its own gold standard cannot calibrate its own claims.

4.1 Two-Tier Architecture and Montage Policy

The screening tier remains the 8-channel Cyton build (Fp1, Fp2, Fz, Cz, C3, C4, Pz, Oz plus reference and ground) at 250 Hz, and it is now formally classified as a screening instrument: it supports band-power estimation, neurofeedback training, and protocol participation, but never source-level claims, never

gamma-band operative features, and never coherence claims beyond the pairs its geometry can defend. The validation tier is a 32-channel actively-driven build on denser geometry, sampled at 500 Hz, with a boundary-element forward model — and it is the only tier where source estimates, operative gamma features, and full coherence maps are legal. The montage comparison below situates both tiers against standard 10-20 clinical subsets, as the review requested; the comparison makes visible exactly which regions the screening tier cannot see, and the framework's estimator map (Chapter 3) was re-drawn under that constraint rather than in ignorance of it.

Region / feature	10-20 clinical routine	Screening tier (8 ch)	Validation tier (32 ch)
Central midline (Cz) SMR	Full	Yes	Yes
Frontal-midline theta (Fz)	Full	Yes	Yes
Parietal alpha (Pz)	Full	Yes	Yes
Anterior frontal (Fp1/Fp2)	Full	Yes	Yes
Lateral temporal (T7/T8)	Full	No — frame limited	Yes (validation geometry)
Occipital (Oz/O1/O2)	Full	Oz only	Yes
Source estimation	Reference standard	Prohibited	Permitted (BEM forward model)
Operative gamma features	Permitted	Prohibited (EMG gate + rate)	Permitted with 500 Hz + gate

Table 5: Capability comparison of the v3.1 tiers against a standard clinical 10-20 subset.

4.2 Electrodes, Contact Quality, and Impedance Policy

v3.0's 50 kΩ dry-electrode ceiling is tightened to 25 kΩ measured at session start, with any channel above threshold excluded from operative features until re-prepped rather than recorded-and-forgiven. The default build moves to active dry or hybrid electrodes with on-electrode buffering, which converts high electrode impedance into manageable amplifier input impedance and materially reduces mains pickup; a saline-gel upgrade path is documented with measured SNR deltas so operators can trade convenience for signal quality knowingly. Contact quality ceases to be a one-time check: the screening tier records continuous impedance telemetry where the board supports it, and every session's quality report logs per-channel impedance at start, at each block boundary, and at end, because the review correctly noted that dry contacts degrade over a session and that degradation is itself data. Real-time adaptive gain, enabled on boards that support it, keeps the ADC in range as offsets drift with contact state.

$$Z_{ch} \le 25\ k\Omega \ \ (start,\ per\ block,\ end);\ \ excluded\ otherwise \tag{4.1}$$

4.3 Sampling, Timing, and Link Budgets

Sampling policy follows the features, not the reverse. The screening tier's 250 Hz remains adequate for bands below 50 Hz by the Nyquist criterion with antialiasing margin, but gamma-band operative features require 500 Hz, which is one reason gamma is exploratory at the screening tier and never operative on the Daisy extension, whose aggregate-rate reduction to 125 Hz caps its legal features at those below 40 Hz. Link policy prefers USB for all trial sessions: BLE is permitted for mobility and ecological sessions only, with packet-loss and jitter quantified per session and gates of at most 0.5% dropped packets and 1 ms RMS clock jitter, measured by sequence-number accounting rather than assumed absent. These budgets are not decorative — Section 5.4's coherence estimators are sensitive to jitter-induced phase noise, and a trial that fails its link budget is invalid regardless of how its spectra look.

$$f_s \geq 2.5 \times f_{max,band};\quad \gamma\text{-operative: } f_s \geq 500\ Hz;\quad \text{Daisy: } f_{legal} < 40\ Hz \tag{4.2}$$

4.4 Auxiliary Channels and Motion Co-registration

Every screening-tier build from this revision reserves two bipolar aux channels for vertical and horizontal EOG and two for jaw and neck EMG, with optional ECG where a third pair is available; the validation tier requires all four plus respiration belt. These channels are not conveniences: the regression pipeline of Section 5.1 uses them as reference regressors for ocular and muscle artifact, and their absence in v3.0 is precisely why that pipeline's ICA had to over-interpret 8 channels of mixed signal. A head IMU is added at both tiers for motion co-registration — its tri-axial record is logged with the EEG stream and used both for artifact flagging and for the anti-gaming rules of Section 6.3, since movement strategies were one of the review's named ways to game a neurofeedback ladder. The IMU clock shares the acquisition clock, so motion and EEG are comparable at millisecond resolution without cross-device alignment.

4.5 Reference Policy and Sensitivity Obligations

v3.0's linked-mastoid reference was flagged for spatial bias and for its particular toxicity to coherence measures, which inherit the reference in every channel pair. v3.1 keeps linked mastoids as the acquisition reference for continuity with the existing build fleet but adds two obligations. First, re-referencing is computed in software and reported side by side: linked-mastoid, average-mastoid, and reference-transformed (REST) versions of every operative feature appear in the session's quality report, and any feature whose sign or rank order changes across references is reference-bound and ineligible for protocol decisions, per Section 3.5. Second, coherence-based features are computed exclusively on re-referenced data, because the review's volume-conduction analysis showed that shared-reference contamination can inflate coherence by amounts comparable to the effects the dyadic protocols seek — an artifact the imaginary coherence of Section 5.4 is designed to remove.

4.6 Calibration and Quality Control

Two calibration procedures are now mandatory and versioned. The bench calibration applies a known signal generator — a 10 µV, 10 Hz sine through a resistor network at each electrode input — before first use and

quarterly thereafter, verifying gain accuracy within 2%, noise floor below 1.5 µV RMS in the 1–40 Hz band, and channel crosstalk below −60 dB; the results are hashed into the session provenance record. The common-mode check applies the generator in common mode to verify rejection above 100 dB with battery isolation, which is the review's point about capacitive coupling: battery power removes ground-loop paths but not capacitive pickup, and only a measured common-mode rejection number tells an operator whether their shielding and cabling work. Thermal and humidity logging at the scalp completes the environment record, since sweat is both an impedance hazard and an EMG generator.

4.7 Ergonomics and Session Environment

The Mark IV frame's pressure variability is managed, not ignored: the build spec adds pressure-relief pads at the four crown contact points, a torque-checked headband fit procedure, and a comfort self-report at block boundaries, because pain is both an artifact source and an ethics issue. Sessions are capped at 90 minutes regardless of protocol, with a wick liner change available at block boundaries in warm conditions, and the temperature-humidity record from Section 4.6 accompanies every session. Long-session thermal drift was one of the review's named failure modes, and the combination of caps, liners, and environment logs addresses it at the level of engineering rather than willpower.

4.8 Bill of Materials, Drawings, and the v3.1 Cost Ladder

The build path is now a versioned artifact: the bill of materials pins part numbers, firmware versions, and board revisions to a repository hash, and the mechanical drawings — including the aux-channel mounting points and IMU bracket — are versioned alongside. The cost ladder below supersedes v3.0's by including consumables, the calibration source, and spares, which the review showed had been understated; the honest ladder is roughly a quarter higher than v3.0's figure, and the framework states it that way. Cost-effectiveness and feasibility metrics accompany each tier so that a prospective operator can see not just what a build costs but what it enables.

Tier	Components included	v3.0 figure	v3.1 figure
S0 — Screening (8 ch)	Cyton + Mark IV + active dry electrodes + aux cables + IMU + liners + calibration source + spares	$640	$810
S1 — Screening + gel path	S0 + saline/gel electrode kit + abrasive prep consumables (12 sessions)	$640	$905
V0 — Validation (32 ch)	Cython + dense active array + BEM-capable workstation + imaging referral budget	—	$4,850
T0 — Trial room	Isolation retrofit (Section 7.5): shielding, acoustic treatment, leakage sensor kit	—	$1,400

Table 6: Cost ladder v3.1 with consumables, calibration, and spares included. v3.0 quoted only S0's first column figure.

5. Signal Processing Pipeline v3.1

The v3.0 pipeline had the right skeleton — filter, artifact-reject, spectral estimate, normalize, estimate features — and the wrong details in almost every stage: Welch windows too short for theta resolution, ICA asked to do more than 8 channels permit, magnitude-squared coherence inflated by volume conduction, z-scores computed against a single baseline state, and no operational cross-frequency estimator despite the theory invoking one. v3.1 rebuilds the pipeline stage by stage under a single constraint: every stage must be containerized, unit-tested, and parameter-recovered before it touches real data. That constraint comes straight from the review, and it is the difference between a pipeline that is documented and one that is reproducible. The full preprocessing provenance is logged in BIDS-EEG format, aligning the framework with the open-neuro ecosystem and its toolboxes (MNE, EEGLAB) through validated wrappers rather than bespoke reimplementations.

5.1 Artifact Removal: ASR Primary, ICA Advisory

Artifact handling is restructured around what the review demonstrated: artifact subspace reconstruction (ASR) on the multi-channel stream is the primary cleaner, using the aux channels of Section 4.4 as regression references for ocular and muscle components before ASR runs; ICA is demoted to advisory status — it may inform the operator about components present, but operative features no longer depend on an 8-channel ICA decomposition whose degrees of freedom cannot separate neural from muscle sources reliably. Online operation uses an ASR-lite variant with fixed calibrated thresholds, because the neurofeedback loop of Section 6.3 cannot afford a data-adaptive method changing behavior mid-session. The cleaning chain is deterministic given its inputs and seed policy, so any session can be replayed bit-for-bit from raw data — a property the containerization section makes enforceable rather than aspirational.

5.2 Spectral Estimation: Multitaper Standard

Spectral estimation switches from Welch periodograms to the multitaper method with a 0.5 Hz resolution floor: 4-second windows, 7-taper Slepian sequences for the standard bands, and wavelet or Stockwell transforms where time-frequency resolution matters more than variance. The review's arithmetic was straightforward — a 2-second Welch window gives 0.5 Hz resolution but with high variance at the theta frequencies where the framework's most important features live, and the frontal-midline theta estimator needs both resolution and stability. Band powers are integrated over the individualized edges of Eq. 3.1, and every spectrum stored in the session record carries its estimator identity, window set, and taper count so that spectra are never compared across incompatible estimators. The aperiodic 1/f component is fitted and archived alongside, both for IAF estimation and because aperiodic drift is itself a vigilance signal the protocols now monitor.

$$\hat{S}(f) = (1/K) \sum_k \left| \sum_t w_k(t)\, x(t)\, e^{-i2\pi f t} \right|^2, \quad K = 7 \text{ tapers}, \Delta f = 0.5 \text{ Hz} \tag{5.1}$$

5.3 Normalization: Multi-State, Robust, Drift-Aware

Normalization replaces the single eyes-open z-score with two changes. First, the baseline is multi-state — eyes-open, eyes-closed, and task blocks from session 0 — and features are z-scored within state, because the review correctly noted that a single baseline conflates state and trait variance. Second, the normalization is robust: centering and scale use the median and 1.4826-scaled MAD rather than mean and standard deviation, which limits the influence of exactly the transient artifacts the cleaning chain cannot fully remove. Within sessions, a drift monitor tracks non-stationarity: when a feature's rolling median moves more than 1.5 robust SDs from its block-1 value, the segment is flagged and renormalized segment-wise, with the flag visible in the session record. Non-stationarity is thus handled by detection and adjustment rather than by the silent assumption that a 90-minute session is one statistical population.

$$z_{robust} = (x - median_{base}) / (1.4826 \times MAD_{base}) \tag{5.2}$$

5.4 Coupling Estimators: Imaginary Coherence and Null Models

Coupling estimation is rebuilt around three estimators with distinct responsibilities. Imaginary coherence replaces magnitude-squared coherence for every inter-site and inter-person claim, because its imaginary part is insensitive to the zero-lag volume-conduction and shared-reference components that physically cannot reflect true interaction; the phase-slope index supplements it as a frequency-resolved direction check. PLV is retained only with an explicit low-SNR bias correction and a minimum-trial gate of 40 clean epochs per condition, since the review documented the upward bias of naive PLV exactly where the framework operates. And every coupling result is evaluated against two null models — phase-randomized surrogates for within-pair structure and a shared-environment null that shuffles across recording pairs to capture common input — because the review's critique of dyadic synchrony was that shared respiration, movement, or room noise can generate coherence that no phase randomization removes. A coupling claim survives only when it beats both nulls.

$$ImC_{xy}(f) = Im[S_{xy}(f)] / \sqrt{[S_{xx}(f) S_{yy}(f)]} \tag{5.3}$$

$$PLV_{corr} = PLV - bias(SNR, N_{trials}), \quad N_{trials} \geq 40 \tag{5.4}$$

5.5 Spatial Filtering and Reference Sensitivity

The pipeline adds a surface-Laplacian stage at the validation tier, where 32 channels make a credible second spatial derivative possible; at the screening tier, a nearest-neighbor approximation is computed for C3-Cz-C4 and Fz-Fp1-Fp2 triangles and used for SMR and frontal-midline features respectively. Spatial filtering sharpens the local specificity of exactly the features the ladder trains, and the review identified its absence as

a principal reason v3.0's site claims were blurred by volume spread. The reference-sensitivity battery of Section 4.5 runs at this stage, since re-referencing is a pipeline transform here: every operative feature is computed under all three reference schemes and the spread is archived with the feature, making reference-boundness a measured property of each result rather than an occasional methodological remark.

5.6 Residual EMG Quantification and the Gamma Gate

The review's most damaging single technical point was that gamma-band features at consumer SNR remain contaminated by residual EMG even after ICA, because scalp EMG power rises steeply above 30 Hz and overlaps the band of interest. v3.1 responds with a quantified gate rather than a hope. The residual EMG index is computed per epoch as the ratio of high-frequency power (80–200 Hz, where EEG contributes little and EMG dominates) projected onto the EMG aux channel's covariance structure, and operative gamma features require the index below 0.4 with the aux-referenced regression residual below threshold — epochs failing the gate are excluded, sessions exceeding a 30% exclusion rate are void for gamma purposes. The gate is deliberately conservative: it will void gamma estimates that a looser standard would keep, and the framework accepts that trade because a contaminated gamma feature is worse than no gamma feature — it validates a proxy that muscle generated.

$$R_{EMG} = P_{80\text{–}200\ Hz}(\ residual \mid EMG\text{-}reg \ ) \ / \ P_{80\text{–}200\ Hz}(\ raw \ ) \ < 0.4 \tag{5.5}$$

5.7 Cross-Frequency Coupling and HCI as Fit Parameters

v3.0 mentioned cross-frequency coupling and listed an alpha–gamma interaction without ever formalizing either. v3.1 operationalizes phase-amplitude coupling with the standard modulation index: the phase of a low-frequency band (theta or IAF-centred alpha) is extracted, the amplitude envelope of a high-frequency band is taken, and the Kullback-Leibler divergence of the amplitude-by-phase distribution from uniformity is the index, evaluated against both shift-predictor and surrogate nulls. The alpha–gamma interaction listed in v3.0 is thereby either formalized as this PAC estimate or removed from the document — there is no third status. The Human-Coupling Index of the dyadic protocols, whose weights the review called arbitrary, is redefined: its weights become fit parameters estimated per cohort under the hierarchical model of Section 3.2, with a sensitivity analysis over the plausible weight range reported alongside every HCI value, so the index's conclusions must survive its own uncertainty.

$$MI_{PAC} = \Sigma_{\varphi}\, p(\varphi)\, \log[\ p(\varphi) \,/\, uniform\ ], \quad p(\varphi) = amplitude\ by\ low\text{-}freq\ phase \tag{5.6}$$

5.8 Containerization, Recovery Tests, and Provenance

The pipeline ships as a version-pinned, signed container with a unit-test suite and two standing verification requirements. Parameter-recovery simulations run before any empirical fit: synthetic data with known

spectral and coupling parameters are passed through the container, and the fitted-versus-true regression must meet pre-set recovery bounds before the container is licensed to process real sessions — a practice the review called for explicitly. BIDS-EEG provenance logging records every transformation from raw file to operative feature, and the analysis-code freeze protocol of Section 8.4 points at this container by hash, so a pre-registration's analysis is executable and immutable. Validated wrappers to MNE and EEGLAB are maintained for operators who bring recordings from other tools, with round-trip tests proving numerical equivalence on the shared test corpus. The pipeline is thus no longer a parameter list in an appendix; it is an artifact with an identity, tests, and a paper trail.

6. Hyper-Correlation Amplification Ladder v3.1

The five-level ladder survives the review with its structure intact and its evidence regime transformed. v3.0 asked operators to climb from SMR training to dyadic coherence with sham control deferred to the second validation gate, no learning model, thresholds fixed by convention, and no defense against gaming the feedback loop. v3.1 makes a yoked sham arm mandatory from the first L1 cohort, attaches an exponential learning curve and a dose-response model to every level, replaces fixed exit thresholds with adaptive per-operator thresholds whose calibration is itself a pre-registered procedure, wires automated stop rules into the feedback loop, and closes the gaming vectors the review enumerated — impedance manipulation and movement strategies among them. The ladder's purpose is unchanged: to train states the D^3 map describes and to measure, under controlled comparison, whether the training transfers. What changes is that every claim of progress now has a control group, a model, and an uncertainty statement attached to it.

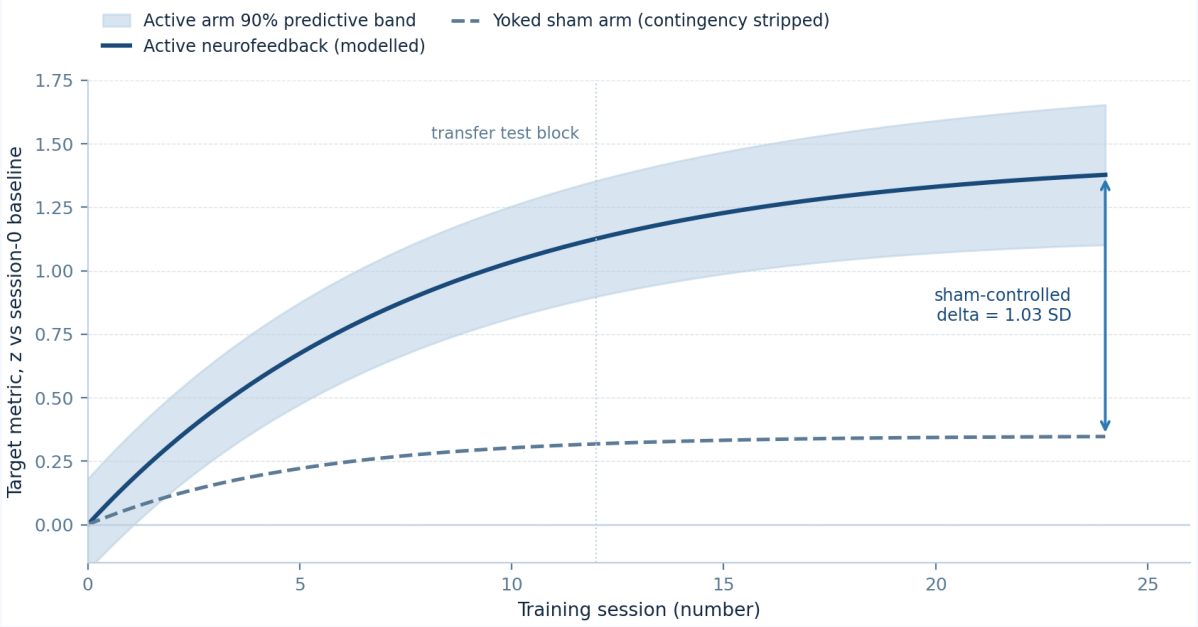


Figure 3: Sham-controlled ladder trajectory model for an L1 SMR cohort. The active arm's curve is a falsifiable model prediction with a 90% predictive band; the dashed sham arm shows the expectancy-and-practice trajectory that active claims must exceed. The transfer test block at session 12 gates level promotion.

6.1 Level Structure Retained

The five levels retain their identities: L1 sensorimotor-rhythm up-training at C3/C4; L2 alpha-theta crossover at Pz; L3 dual-operator coupled sessions; L4 dyadic blind protocols; L5 independent blind-trial operation. What the revision changes is the rules of passage. Promotion now requires a sham-controlled delta, a transfer test block, and a learning-curve fit with a positive learning rate — three conditions instead of v3.0's single threshold crossing. The levels remain cumulative in skills but independent in claims: nothing a later level does retroactively validates an earlier level's model, and each level's exit criteria are calibrated on its own cohort data. The ladder is therefore a sequence of controlled experiments, not a curriculum with an implied guarantee — which is exactly the epistemic status the review said the evidence could support.

6.2 Sham Control and Expectancy Measurement

Every training cohort from this revision runs a yoked sham arm: participants receive feedback drawn from a matched partner's session or from their own prior session's recording, time-locked to the session structure so that the contingency between their brain state and the feedback is stripped while every other cue — screen, sounds, session script, experimenter behavior — is preserved. Yoked sham is chosen over random feedback because it preserves the temporal statistics of real feedback, denying participants the simple contingency-detection cue that random signals provide. Expectancy and demand characteristics are measured, not assumed away: a pre-registered questionnaire battery is administered at intake, mid-ladder, and exit, and its scores are covariates in every cohort analysis. The review was explicit that practice and expectancy effects can produce the entirety of a neurofeedback result; the sham arm and the battery are the framework's acceptance of that burden of proof.

6.3 The Real-Time Loop: Latency, Stops, and Anti-Gaming

The feedback loop is specified as an engineered system with a 300-millisecond end-to-end latency budget from neural event to feedback onset, measured with a test-signal harness and logged per session; contingency quality is verified against surrogate feedback at calibration time, because a loop whose contingency cannot be distinguished from yoked feedback is not a training instrument. Online artifact rejection uses the ASR-lite chain of Section 5.1 with excursion gating, and epochs passing artifact criteria feed a CUSUM-based automated stop rule: when the standardized cumulative sum of the target metric's divergence from its session trend crosses the control-band boundary, the session terminates automatically — the review's demand for real-time stop rules answered with a named algorithm and a logged trigger. Anti-gaming closes the loop: impedance excursions, IMU-measured head or jaw movement, and EMG-channel activation all invalidate the corresponding epochs for scoring, so the strategies the review named for climbing the ladder spuriously are detected and costed rather than punished by suspicion.

$$CUSUM_t = max[\,0,\, CUSUM_{t-1} + (\,z_t - \mu_0\,) / \sigma_0 - k\,], \quad stop\ at\ CUSUM > h \tag{6.1}$$

6.4 Alpha-Theta Repositioned

L2's alpha-theta crossover protocol is repositioned honestly. The review found v3.0's boundary-state language unsupported: the specific claims about hypnagogic boundary states and their significance rest on a literature that does not reliably beat sham controls. v3.1 therefore retains alpha-theta as a relaxation and affect-regulation protocol with a well-defined training target — increased theta fraction with maintained alpha — and deletes the boundary-state interpretive layer from operative text. The dissociation risk the review documented is answered in the safety chapter with screening, paired-operator rules, and stop criteria; within the ladder, L2's exit criteria are recalibrated to the relaxation target, and the level's contribution to later levels is framed as state-flexibility training rather than gateway mysticism. The framework loses a dramatic narrative and gains a defensible one.

6.5 Adaptive Thresholds, Transfer, and Learning Models

Exit criteria become individualized and model-based. Each operator's reward threshold follows the adaptive rule of Eq. 6.2: it starts at their robust baseline plus k standard deviations and adjusts with a bounded step as training progresses, keeping success rates in the 40–60% band that the learning literature identifies as training-productive; floors and ceilings prevent the rule from drifting into triviality or impossibility. Level promotion requires the transfer test block: performance measured on sessions and metrics that were not part of the feedback loop, because the review noted that sequential leveling assumed a transfer v3.0 never demonstrated. The learning-curve model of Eq. 6.3 fits an exponential approach to each operator's cohort-adjusted metric, yielding a per-operator learning rate with a confidence interval — the quantitative learning measure v3.0 lacked — and the dose-response counterpart relates cumulative training minutes to retention at 30 and 90 days. Responder definitions are pre-specified at cohort registration: a responder is an operator whose sham-controlled delta exceeds the cohort's 60th percentile with a CI excluding zero, a definition frozen before data collection rather than negotiated after it.

$$thr_{n+1} = thr_n + \delta \cdot sign(\, rate_n - 0.5\,),\ \ \delta \text{ bounded, } thr \in [\, \mu_0{+}0.5\sigma,\ \mu_0{+}2.0\sigma\,] \tag{6.2}$$

$$perf(\,n\,) = a - b \cdot e^{-n/\tau},\quad \text{learning rate } \lambda_L = b \,/\, \tau \text{ (CI by bootstrap)} \tag{6.3}$$

6.6 Load, Recovery, and the Allostatic Anchor

The load integral of Eq. 6.1 in v3.0 was a heuristic without a physiological anchor, and the review said so plainly. v3.1 keeps the integral as the accounting device but anchors it where hardware permits: when an ECG channel is present, the load weights incorporate an HRV-derived index (RMSSD deviation from the operator's baseline), and the weekly cumulative cap — new in this revision — bounds total load across sessions, not just within one, addressing the allostatic accumulation the review flagged. Recovery blocks cease to be subjective: a standardized checklist plus validated short-form mood and dissociation scales are administered before session close, their scores logged, and any score outside the operator's historical band triggers the safety chapter's review path. The operator-logged lucidity rating of v3.0 is replaced by the

standardized post-session report instrument, which the dashboard can hash and archive — verifiable structure instead of unverifiable introspection.

$$\Lambda_{week} = \Sigma_{sessions} \int_0^T w_{HRV}(t) \cdot |\Delta P(t)| \, dt \le \Lambda_{cap} \tag{6.4}$$

6.7 Dyadic Protocols under Modern Hyperscanning Practice

L3 and L4 inherit the review's hardest objection: inter-person coherence can be manufactured by shared respiration, synchronized movement, common auditory input, or a shared reference, and v3.1's response is a control stack rather than a disclaimer. Every dyadic session logs respiration belts and IMU streams at both operators; coupling estimates are the imaginary coherence of Section 5.4, computed after partialling the shared respiratory and motion covariates; both surrogate nulls of Section 5.4 are applied; and an inter-dyad control arm — pairs recorded in matched sessions but never interacting — provides the between-pair baseline that within-dyad crossover designs cannot. The level's claims are restricted accordingly: what L3/L4 can establish, at best, is coupling beyond these controls under blind conditions, and the EP-3 gate (Chapter 8) was recalibrated to the effect-size range the published hyperscanning literature actually reports. Alignment with contemporary hyperscanning best-practice guidance — synchronized clocks with drift budgets, pre-registered dyad assignment, and disclosure of all pairing metadata — is adopted in full.

7. Channel and Remote-View Protocol v3.1

The trial layer survives with its expected-null clause intact and its engineering rebuilt from the foundation up. The review's verdict on this layer was double-edged and the framework accepts both edges: the historical effect sizes for remote-viewing and ganzfeld performance do collapse under strict pre-registration and blinding, and simultaneously the v3.0 protocol left too many leakage paths — sensory, temporal, statistical, and social — unquantified for its own blind results to mean what it wanted them to mean. v3.1 therefore rescores the layer's purpose: it is resourced as a negative-control program with capped budget and pre-committed publication symmetry, run at a level of isolation engineering that makes a positive result informative if one ever survives. Priors are reset to a conservative range around zero; the minimal effect worth detecting is declared before data collection; and a positive-control arm, measuring how far subtle cues alone can move scores, becomes a primary endpoint rather than a methodological afterthought.

7.1 Priors, Expected Null, and the Information Bound

Three commitments frame the layer. First, the prior on anomalous reception is set near zero and stated numerically: the Bayesian sequencing of Section 8.2 uses a prior centered on zero effect with a width that makes even the upper tail of the historical database generous. Second, the expected-null clause is retained: the design anticipates a null outcome and treats it as the informative result it is, scoped per Section 8.6 to the specific configuration tested. Third, the layer states its information-theoretic frame: a four-alternative forced-choice trial carries at most two bits, any true transfer mechanism must respect the data-processing

inequality — no post-hoc processing of recordings can add information the trial did not contain — and the minimal effect of interest is set at 8% above chance ($p_0 = 0.25$ vs. a detectable $p_1 = 0.33$), with $q = 1 - p$. Run as part of this revision, the power analysis of Eq. 7.1 delivered an uncomfortable result the framework has adopted rather than buried: v3.0's registered N = 100 carries only about 50% power for that minimal effect at one-sided $\alpha = 0.05$, so v3.1 doubles the registration to N = 200 — the size that delivers 80% power. The registered N is thus derived, not asserted — the review's demand that N be justified by power analysis rather than tradition, answered with an analysis that corrected the framework's own prior registration.

$$C_{4AFC} = \log_2(4) = 2 \text{ bits/trial};\ N = [\, z_\alpha \sqrt{(p_0 q_0)} + z_\beta \sqrt{(p_1 q_1)}\,]^2 / (p_1 - p_0)^2 = 192, \text{ register } N = 200 \tag{7.1}$$

7.2 Target Selection and Randomization

Target-pool design closes the guessing strategies that semantic clustering permits. The pool is constructed with a semantic-distance floor: any two targets in the active pool must exceed a pre-computed embedding-distance threshold, so that even a receiver who correctly recovers the pool's neighborhood cannot gain differential advantage across candidates. Randomization uses a certified hardware RNG whose output is seeded publicly — the seed committed via external timestamp before the session window — and stratified so that target position, sensory category, and sender identity are counterbalanced across the sequence. Every randomization event is hashed into the audit chain with its inputs and outputs, making the full target sequence reconstructible and auditable after unsealing. The review's complaint that cryptographic sealing does not address sensory leakage is correct and is answered in Section 7.5; what sealing does address — post-hoc target selection and silent re-draws — is here made impossible rather than merely discouraged.

7.3 Judging: Automated Features Primary, Witnesses Secondary

Free-response scoring is rebuilt around a pre-registered automated feature vector: twelve features — lexical-field overlap, embedding-similarity rank, structural-sketch descriptors, color-term congruence, and eight domain-specific counts — are computed by frozen code before any human sees the transcripts, and they constitute the primary endpoint. Human judging is secondary and consensus-structured: a pool of six blind witnesses, recruited outside the operator community with conflicts declared, rank transcripts against targets, pairwise agreement is reported (Krippendorff's α), and the witness-derived endpoint enters the family-wise correction of Section 8.3 as a secondary family. The witness pool is expanded from v3.0's three because three was minimal for free-response reliability, and small-community independence is hard to guarantee — six external judges with declared conflicts and published agreement statistics is the floor the review accepted. Post-hoc rationalization of sketches is blunted by procedure: transcripts and sketches are hashed and timestamped before feedback, and scoring runs only on sealed artifacts.

7.4 Sealing, Blinding, and Session SOP

The blinding standard is now a written SOP with named roles. Allocation to sender/receiver/senderless conditions is sealed by the audit chain and revealed only after scoring; experimenters who interact with receivers are themselves blind to target identity; decoy sessions — structurally identical sessions with no target — are interleaved at a 1:4 ratio so that neither session rhythm nor experimenter behavior carries target information; and dyadic participants are blind to condition, hypothesis, and even to the existence of the senderless arms, closing the v3.0 gap the review identified in L4 blinding. Session artifacts — recordings, transcripts, sketches, judge worksheets — are hashed at creation and again at scoring close; any hash mismatch voids the session. The SOP's purpose is not theater: each control maps to a specific leak the review documented in historical programs, and the SOP makes each leak either impossible or measurable.

7.5 Quantified Sensory Isolation and Leakage Models

v3.0 asserted sensory isolation; v3.1 specifies it numerically and measures it. The trial room's isolation envelope is defined by the table below, verified at commissioning and re-verified quarterly with calibrated instruments — an EM field probe, a sound-level meter, an airflow anemometer, and scheduled olfactory controls. Beyond the envelope, a quantitative leakage model assigns each channel (sound, timing, thermal, prosody, micro-expression) an estimated information capacity given the room's measurements, so that the total leakage budget of the room is a computed number the protocol can compare against the minimal effect of interest. Red-team exercises — Section 8.5 — periodically attempt to extract target information through the room as built, and the attempts and their results are published in the errata register. Isolation thus ceases to be a claim and becomes an instrumented, adversarially tested property of the facility.

Channel	Specification (envelope)	Verification instrument
Electromagnetic	Ambient 50/60 Hz field ≤ 5 mG in receiver position; no wireless emitters within 3 m	EM field probe, annual + quarterly spot
Acoustic	Background ≤ 25 dB(A); sender room airborne isolation STC ≥ 50; no mechanical coupling of HVAC	Sound-level meter, commissioning + quarterly
Airflow / thermal	Inter-room airflow ≤ 0.05 m/s at receiver position; ΔT between rooms logged < 1.5 °C	Anemometer + differential logging
Olfactory	No food/scents protocol; positive air pressure gradient from receiver room	Scheduled scent controls
Timing	No session-schedule regularity exploitable to infer target class; randomized session lengths	Schedule audit against sequence

Table 7: Quantified isolation envelope for the trial room.

7.6 Timing and Synchronization Budgets

The lab streaming layer remains the transport for multi-device sessions, with two additions that answer the review's clock-drift objection. Sync markers — a shared click train — are injected at session start, at each block boundary, and at session end, and inter-device drift is measured against the markers with a budget of at

most 1 ms RMS across a session; sessions exceeding budget are flagged and their cross-device timing claims downgraded. Where budgets matter most — the dual-EEG dyadic sessions of Chapter 6 — the clocks are disciplined to a common hardware trigger rather than software estimates alone. The drift budget converts a silent failure mode into a measured, archived quantity, consistent with the framework-wide rule that every quantity that can silently corrupt a claim must either be measured or be bounded.

7.7 The Positive-Control Arm as Primary Endpoint

The review's most constructive single demand on this layer is adopted wholesale: subtle-cue performance becomes a measured positive control, run as a condition of the trial program with equal design care. In the positive-control arm, a sender is permitted — scripted — to exhibit calibrated micro-expressions and prosodic cues while the isolation envelope is deliberately relaxed along one measured channel; camera and audio capture at the receiver quantify the cue dose delivered, and the receiver's scoring performance under known cue dose calibrates the entire facility's leakage model. The positive-control effect size is a primary endpoint of the program: if calibrated cues move scores by amounts comparable to historical positive results, the burden shifts decisively to any claimed above-cue performance, and if they do not, the leakage model is revised against measurement. Either way the program learns something real — which is more than an uncontrolled blind layer can promise.

7.8 Channel-Entry Bands, Reclassified

The v3.0 entry criteria — a Boundary-axis estimate in the −1.5 to −0.5 band, plus support conditions — were unvalidated bands worn as gates. v3.1 reclassifies them as hypotheses with calibration blocks: each trial program includes per-operator calibration sessions that measure whether the band's occupancy predicts any scoring-relevant property at all, and the band is retuned or retired per operator on that evidence. The twenty-minute session cap is retained as a safety default with a documented titration protocol (Section 9.4), and the twenty-minute figure itself is now marked as a conservative safety choice rather than an attentional law. Entry criteria therefore retain a protocol role but lose their evidentiary pretense; they select sessions, they do not validate states.

8. Validation Gates and Statistics v3.1

The validation program expands from four gates to six and rebuilds its statistical spine around the reporting standards the review invoked. The changes are systematic rather than cosmetic: every gate is now conditional on propagated confidence intervals, null claims require equivalence testing rather than failure to reach significance, aggregation corrects for dependence instead of assuming independence, registration is externally timestamped and code-frozen, and a multi-lab replication gate stands between internal results and any external claim. Two new gates broaden coverage: EP-5 requires independent replication before any estimator or protocol is treated as operative, and EP-6 requires an adversarial red-team exercise to attack the trial layer's isolation and scoring. The program's epistemics are unchanged — kill criteria remain, expected-null stances remain — but its statistics now match its rhetoric.

8.1 The Six Gates

Gate	Question	v3.0 criterion	v3.1 criterion
EP-1	Are estimator features reliable enough to carry protocol decisions?	ICC ≥ 0.6 at 2 weeks	ICC ≥ 0.4 with CI for use; ≥ 0.6 for promotable claims; 8-week drift checkpoint added
EP-2	Does neurofeedback produce sham-controlled gains?	d ≥ 0.4 vs baseline	d ≥ 0.4 vs yoked sham on a multiplicity-protected primary; expectancy covariates included
EP-3	Does dyadic coupling exceed controls?	PLV lift ≥ 2 SD	ImCoh lift recalibrated to published hyperscanning ranges, vs inter-dyad control + both nulls
EP-4	Does the trial layer beat chance under full isolation?	One-sided z ≥ 1.64 at N=100	Dual gate: z ≥ 1.64 triggers; claim requires effect-size CI exceeding equivalence margin and surviving positive-control comparison
EP-5	Do results replicate outside the originating lab?	— (did not exist)	Multi-lab replication (≥ 2 independent sites, shared protocol container) required for operative status
EP-6	Can the isolation and scoring be defeated by adversaries?	— (did not exist)	Red-team exercise with published attempts, success rates, and remediation

Table 8: The validation gates of v3.1.

8.2 Statistical Machinery: Intervals, Equivalence, Bayes

Every results table in every gate now carries confidence intervals computed with the propagated variance of Section 2.4 and block bootstrap (B = 2000, seed policy pinned) — the review's under-specified bootstrap is here fully specified. Null claims are stated as equivalence results: the two-one-sided-tests (TOST) procedure of Eq. 8.1, with equivalence margins declared in the pre-registration, so that a null is an affirmative finding about an interval rather than an absence of significance. Bayesian sequential analysis runs in parallel with pre-specified priors and a Bayes-factor reporting rule, giving the program a second lens that does not share frequentist assumptions. Analyses model site and cohort as random effects; competing pre-registered estimator forms are compared by WAIC/LOO when both were registered; and measurement-invariance tests across operator subgroups verify that the feature-to-axis mapping behaves identically across the populations the framework serves. The naive Stouffer aggregation of v3.0 is retired: aggregation across sessions uses the corrected-variance form of Eq. 8.2, which down-weights correlated contributions, or block bootstrap where the correlation structure is too irregular to model.

$$\text{TOST: } H_0\text{: } |\delta| \geq \Delta_0 \text{ rejected if } CI_{90}(\delta) \subset (-\Delta_0, +\Delta_0) \tag{8.1}$$

$$Z_{corr} = \Sigma w_i z_i \,/\, \sqrt{[\,\Sigma w_i^2 + 2\,\Sigma_{i<j} w_i w_j \rho_{ij}\,]},\ \rho \text{ from block bootstrap} \tag{8.2}$$

8.3 Families, Multiplicity, and Covariate Structure

Multiplicity control becomes family-explicit. Each gate pre-registers its family of tests — the full set of contrasts that could support its claim, not a favorable subset — and applies a hierarchical false-familywise policy: the gate's primary endpoint carries the family's alpha; secondaries are tested only under a gatekeeping scheme that spends no additional type-I error on unguarded tests. The free-response and forced-choice endpoints of the trial layer form one composite family with the correction allocated in pre-registration, answering the review's multiple-testing finding across endpoint families. Order and fatigue effects are pre-registered covariates with counterbalanced designs specified in the SOP; a result that flips under order counterbalancing is reported as order-bound and ineligible for claims. The full family definition per gate is part of the frozen analysis code, so multiplicity protection is verified by artifact rather than by trust.

8.4 Registration, Timestamps, Code Freeze, and Interim Rules

The registration stack is hardened end-to-end. Pre-registrations are posted to a public registry (OSF or equivalent) and anchored externally with RFC-3161 timestamp tokens, so that priority no longer depends on the integrity of any local clock — the audit hash chain's weakness the review identified. Analysis code is frozen before first data collection: the exact container hash (Section 5.8) is recorded in the registration, deviations require a documented amendment before unblinding, and the amendment log itself is timestamped. Interim analysis is permitted exactly once per gate under a pre-specified alpha-spending function, with the interim look's existence and timing itself registered; unregistered looks are protocol violations that void the gate. Optional stopping is unreachable by construction: the sequence length is fixed in the registration, the audit chain records every session as it completes, and a gate's trial is terminated — not continued — by any deviation. These rules make the registration verifiable by third parties rather than merely visible to them.

8.5 Open Science Provisions and Adversarial Collaboration

The framework adopts the open-science set wholesale, with the consent constraints of Section 9.5 respected. Registered-report format is the required path for EP-gate publications; CONSORT-style participant flow diagrams accompany every experimental arm; negative results are published with the same prominence and the same channel as positive ones, a commitment backed by the errata register; raw data and code are released under open licenses where consent allows, and their absence for consent reasons is documented rather than silently gaps. Adversarial collaboration is institutionalized: before any gate that could flip a framework position, an independent analysis team with opposing priors is invited to co-design the analysis, and their sign-off is part of the gate record. The red-team provision of EP-6 extends the same posture to the infrastructure: isolation and scoring are attacked periodically by parties incentivized to succeed, and the results — successes included — are published. The review's warning that a small community amplifies confirmation bias is met with structure, not sentiment.

8.6 Retirement, Retraction, and the Living Registry

Failure handling is made explicit. When a gate fails, the affected claim is retired with a scope statement: retirement applies to the specific configuration — hardware tier, estimator version, protocol parameters — that failed, and to no other; the retirement flag propagates to the dashboard (Chapter 10), which marks the affected estimates as retired and refuses them to protocol decisions; and informal continued use is addressed by telemetry note rather than pretense — the framework cannot police private practice, but it can refuse retired claims its own infrastructure's endorsement. A retraction and revision policy governs erroneous publications symmetrically with the errata register. The living prediction registry closes the loop opened by the review: every operative claim, its evidence gate, its replication status, and its retirement scope are recorded in a versioned register whose hashes join the timestamp chain, making the framework's epistemic state inspectable at any moment.

9. Safety, Ethics and Governance v3.1

The safety chapter absorbs the review's consistent theme that v3.0's protections were real but shallow: self-report where validated instruments exist, single-session caps where cumulative load matters, disclaimers where governance is needed. v3.1 deepens each layer. Screening and referral are structured with named pathways and documented triggers. Dissociation safeguards around alpha-theta work become instrumented. Consent becomes granular and revocable. Adverse events acquire a definition, an instrument, an independent monitor for multi-operator programs, and a clinical-review step before any level's permanent retirement. Data governance gains retention schedules and deletion workflows. And the commercial-prohibition clause gains an enforcement posture it previously lacked. None of this converts a DIY framework into a clinical one — the scope language below is if anything stricter — but it closes the distance between what the document promises and what its procedures deliver.

9.1 Scope, Screening, and Referral Pathways

The self-screening instrument is retained and strengthened with the explicit statement the review demanded: it is not a clinical assessment, and its purpose is referral, not clearance. Positive screens for seizure history, psychiatric instability, dissociative symptoms, or current crisis route to documented referral pathways with named local resources; the framework's protocols are unavailable to screened-positive operators until a licensed professional documents otherwise. Participants aged 18–21 operate under a young-adult protocol requiring guardian consultation for L3+ levels, acknowledging the vulnerability gradient the review identified while keeping the framework's exclusion of minors absolute. Any human data collection intended for external claims requires ethics review — an IRB or recognized equivalent — and internal-only programs must document their self-assessment against the same review criteria, making the gap between the two visible rather than assumed empty.

9.2 Dissociation Safeguards for Alpha-Theta Work

L2's protocol carries the framework's most documented psychological risk, and v3.1 treats it with matching specificity. Screening uses the Dissociative Experiences Scale short form at intake, with scores above the

validated clinical cutoff excluding operators from L2 absent professional guidance. In-session monitoring pairs the standardized recovery scales with operator behavioral signs, and the automated stop rules of Section 6.3 include dissociation-signature triggers — sustained theta fraction above the training band with non-response to the closure script among them. The paired-operator rule requires a second present adult for all L2 sessions, both as safety presence and as post-session reality-orientation support. The recovery block includes a grounding script and a structured re-entry check whose scores are logged; operators whose recovery scores leave their historical band twice in any rolling month are suspended from L2 pending the review path. These are the review's safeguards, adopted and instrumented rather than restated.

9.3 Covariates, Exclusions, and Substance Windows

Session records now carry the covariate set the review specified: hours of sleep (self-report), a validated short-form mood scale, caffeine dose and timing, and a medication and substance log whose exclusions extend beyond v3.0's 24-hour stimulant window — stimulant exclusion is 72 hours for operative sessions, with residual-effect acknowledgment that some agents (long-acting compounds, antidepressant titration periods) require a documented washout conversation rather than a fixed clock. Actigraphy-based sleep verification is optional at the screening tier and required at the validation tier, because self-reported sleep is a known-biased covariate. These covariates are not decoration: the hierarchical models of Chapter 3 and the cohort analyses of Chapter 8 include them, and the invariance tests can detect when an unmodeled covariate is masquerading as a state change — exactly the confound class the review warned the ladder was exposed to.

9.4 Caps: Session, Entry, and Weekly Load

The cap structure is now three-layered. Session caps remain at 90 minutes hard. Trial-layer entry sessions remain capped at 20 minutes by default, with a titration protocol that may adjust within a documented 15–30 minute range based on operator tolerance records — the arbitrary-constant criticism answered by making the constant adjustable and the adjustment rule explicit rather than by pretending the number was derived. The weekly cumulative load cap of Eq. 6.4 bounds total allostatic exposure across sessions, with the cap value set from the operator's first-week baseline and reviewed monthly; exceeding the weekly cap suspends protocol participation until recovery criteria are met. The three layers together answer the review's point that within-session caps alone leave cumulative load unbounded — the allostatic argument the review made and the framework now implements.

9.5 Data Governance: Consent, Anonymity, Retention

Consent becomes granular, tiered, and revocable: each operator's record carries per-dataset sharing tiers (private, cohort-prior, research-release, open-release), each tier separately consented, and revocation propagates to derived artifacts at the next scheduled re-fit of the cohort prior. Anonymization is engineered against re-identification: node IDs are salted with per-release salts, shared bundles meet a k-anonymity floor of $k = 8$ on all quasi-identifier combinations, and aggregate releases introduce calibrated noise per a stated differential-privacy budget. Retention and deletion are scheduled rather than indefinite: raw EEG at 10 years, derived features at 15, audit records at the framework's operational lifetime, with a deletion workflow

that removes an operator's data from every tier within 90 days of request and logs the deletion itself — the log entry being the deletion's only residue. These provisions answer the review's re-identification and retention findings directly, and they apply to the dashboard integrations of Chapter 10 without exception.

9.6 Adverse Events, Monitoring, and System Integrity

Adverse events acquire infrastructure. The definition is operational — any physical or psychological distress attributed by the operator or observer to participation, regardless of perceived severity — and the reporting instrument is structured, timestamped, and hashed into the audit chain; rates are published in the errata register annually. For multi-operator programs an independent safety monitor reviews reports monthly with authority to suspend levels; single-operator programs document their self-monitoring plan as part of the safety charter. The two-event permanent-retirement rule of v3.0 is qualified: retirement now follows clinical review of the events' relation to the protocol, with permanent retirement reserved for related events and conditional retirement available otherwise — the review's point being that an automatic rule can retire a safe level for unrelated causes. Equipment failure and data corruption acquire their own protocol: acquisition checksums, failover recorder for trial sessions, and corruption-handling rules that quarantine affected data rather than reconstruct it. A 90-day post-participation follow-up closes each operator's engagement record.

9.7 Commercial Framing and Enforcement

The prohibition on commercial psychic framing gains teeth. All framework artifacts — the specification, the pipeline container, the dashboard — carry license headers that prohibit commercial invocation of the framework's name for psychic, paranormal, or divinatory services while the framework operates in falsifiable mode; the community enforcement note deputizes the errata register, where violations are logged, and the dashboard's claims firewall (Section 10.3) refuses to display framework-derived estimates alongside any commercial framing. The review's observation that disclaimers do not prevent over-interpretation is accepted as the design constraint it is: the framework cannot control what people say about it, but it can refuse to lend its artifacts' credibility to commercialization, and from this revision it structurally does.

10. Dashboard Integration and Demand Control

The PSP-144 Sovereign Hub integration carries over with three structural changes that answer the review's dashboard findings: the session schema is versioned to v2 with uncertainty, provenance, and retirement fields as first-class citizens; the display policy becomes uncertainty-first, demoting point estimates that v3.0 rendered with false precision; and a claims firewall separates the research instrument from any public-facing surface, with masked-mode operation to control the demand characteristics that live estimates create. The dashboard is the framework's most consequential social interface — it is where estimates acquire the authority of numbers on a screen — and the revision treats it as an experimental instrument with its own bias channels rather than as a neutral window onto the pipeline.

10.1 Integration Architecture and Session Schema v2

The transport remains REST over the local hub with the same endpoint shape operators already know, but the payload schema is versioned to v2 and the version is explicit in every record. The v2 schema adds four field groups the review asked for: uncertainty objects carrying the propagated intervals of Section 2.4 for every estimate; provenance objects carrying the container hash, pipeline version, and calibration record that produced the data; status objects carrying the operative, probationary, exploratory, and retired classifications of Chapters 3 and 8; and consent echoes confirming the sharing tier in force at write time. The schema below shows the v2 session record; the full machine-readable schema ships with the pipeline container and is validated on every POST, so a hub that cannot enforce the v2 contract cannot consume v3.1 sessions.

Session JSON schema v2 — uncertainty, provenance, status as first-class fields

```
POST /api/v2/sessions
{
  "schema": "epf-session/2.0",
  "session_id": "ses_20261019_094231_a41f",
  "operator": "node_id (salted, release-scoped)",
  "acquisition": {
    "tier": "S0",
    "montage": ["Fp1","Fp2","Fz","Cz","C3","C4","Pz","Oz"],
    "fs_hz": 250,
    "impedance": {"start": "...", "blocks": ["..."], "end": "..."},
    "link": {"transport": "usb", "drop_pct": 0.02, "jitter_ms": 0.4},
    "aux": ["eog_v","eog_h","emg_jaw","emg_neck","imu_6axis"]
  },
  "estimates": [
    {
      "feature": "SMR_IAFplus4to7",
      "value": 0.83,
      "ci90": [0.61, 1.04],
      "status": "probationary",
      "provenance": {"container": "sha256:9f41c...",
                     "pipeline": "epf-pipe/1.4.2"}
    }
  ],
  "gates": {"resid_xi_ratio": 0.21, "remg": 0.33, "drift_flags": []},
  "retired_claims": [],
  "consent_tier": "cohort-prior"
}
```

10.2 Uncertainty-First Display Policy

The display policy inverts v3.0's. Point estimates are rendered as ranges first — a band with the interval, the point marked within it — and any estimate whose interval spans the protocol-relevant threshold renders with an explicit indeterminate flag rather than a value. Status badges travel with every tile: operative, probationary, exploratory, retired, with retired tiles visually inert and excluded from any aggregate the hub computes. Trend displays carry their uncertainty forward — a trend line with no interval is a schema violation, not a style choice. The review's false-precision finding was ultimately a finding about persuasive design: a number displayed without its uncertainty invites decisions the data cannot support, and the v2 display policy makes the supported decision set the visible one.

10.3 The Claims Firewall and Demand Control

The firewall has two walls. The first separates research dashboards from public surfaces: framework-derived estimates may not appear on any public-facing rendering, any commercial surface (Section 9.7), or any context lacking the charter caveat, and the hub's API enforces this by refusing non-research client identities the estimate fields entirely. The second wall is temporal: masked-mode operation, in which the dashboard suppresses live feedback of the estimators under test, is the required mode for any session that feeds a validation gate — the operator trains and performs while the estimates accumulate invisibly, which removes the demand-characteristics channel that live displays create. Every masked-mode session closes with the standardized debrief instrument that asks what the operator believed the session was testing; belief reports are covariates in the gate analyses. The hub therefore participates in the framework's epistemics rather than merely displaying their outputs.

11. Conceptual Reconciliation and Non-Claims

The review's conceptual section was the hardest to answer because most of its findings were true in ways that engineering cannot fix: the D^3 axes remain theoretical constructs; the inherited lineage was cited as if established; the healthy reference was arbitrary; the framework simultaneously treated historical psi programs as informative and predicted their nullification. The response is a chapter of honest accounting. This chapter states exactly what the framework's concepts are, what status each inherited layer holds, what the term hyper-correlation now means and does not mean, how the prior tension is handled, and what the framework claims it cannot do. Nothing in the framework's operative machinery depends on any statement in this chapter — which is precisely the point: after v3.1, the operative machinery must survive on its own evidence.

11.1 Status of the D^3 Axes

The Precision, Boundary, and Temporal axes are theoretical constructs of the framework's own lineage, and every EEG mapping in Chapter 3 is correlational. The revised map states this per estimator rather than globally: each feature row in Table 4 carries its status, each status carries its evidentiary gate, and the estimators' predictive-coding correspondences of Section 2.7 are stated as mappings under test — constructs that could in principle be refuted by the invariance and sensitivity batteries. The reception posture of v3.0, which the review identified as a post-hoc label for a state region, is relabeled the operator-selected posture window: a session-scheduling parameter with no inferential role, stripped of its interpretive costume. The axes remain the framework's organizing hypothesis; v3.1 simply stops charging that hypothesis to the evidence account before the evidence arrives.

11.2 The Inherited Layers, Demoted

The lineage table of v3.0 is revised: UTD v0.3's axis algebra remains the state-space definition — explicitly reused as a modeling commitment rather than an empirical result; the comorbidity distance metric and its

atlas coordinates are marked interpretive with no operative use anywhere in v3.1's protocols; the Coherence Polytope and UHG invertibility bounds are acknowledged as non-operative and moved to the lineage appendix, referenced as intellectual ancestry rather than load-bearing structure. The review's finding — that v3.0's version charter reused non-empirical layers as if they were established — is resolved by this demotion, which costs the framework nothing operative and buys it consistency: no claim in v3.1 rests on a layer the framework itself has not tested. The plasticity noise that v3.0 left symbolic was operationalized in Section 2.5; the game-theoretic and mesh extensions of prior work remain explicitly out of the EEG layer's scope, deferred with the backlog's trigger conditions rather than gestured at.

11.3 The Hyper-Correlation Fork

The review observed that v3.0 redefined hyper-correlation as subtle-cue performance, quietly collapsing the original claim while preserving the term's rhetorical charge. v3.1 formalizes the fork the observation implies. HC-S (subtle-cue hyper-correlation) denotes above-chance performance explained by sensory leakage, statistical anticipation, and inference — the computationally real phenomenon the framework's positive-control arm is designed to measure and bound. HC-P (coupling hyper-correlation) denotes inter-person physiological coupling beyond the control stack of Section 6.7 — the phenomenon the dyadic gates test. The terms never substitute for one another: every protocol chapter names which construct it measures, and any text that blends them is a defect reportable to the errata register. The umbrella term hyper-correlation survives only as lineage vocabulary; it carries no claim by itself.

11.4 The Prior Tension, Documented

The framework's posture toward historical psi research contains a genuine tension the review named: it treats those programs as informative enough to demand a trial layer, conservative enough to predict their nullification. v3.1 documents the tension rather than resolving it by fiat. The trial layer exists because the historical database is large enough that a well-instrumented null is itself a contribution to a contested record; the priors are near-zero because that database's effect sizes collapse under modern blinding and registration standards; and the positive-control arm exists because the most likely true story — subtle-cue transfer — is worth measuring in its own right. The tension is thus paid for in design currency: the layer's cost is capped, its publication symmetry is pre-committed, and its expected outcome is written down before the data arrive. A framework that cannot hold this tension honestly should not run the experiments at all; v3.1's position is that it can, and Chapter 12's ledger is the running account of whether it does.

11.5 External Validity Roadmap

The review's closing verdict — that the architecture prioritized internal consistency over external validity — is accepted as the roadmap's starting line. The near-term integrations that raise external validity are specified: HRV multimodal fusion is operative in this revision (Sections 6.6 and 9.3); eye-tracking joins at the validation tier where hardware supports it; the computational-psychiatry instrument batteries that share parameters with the predictive-coding mapping are listed for the validation cohort; network-topology metrics for dyad pools join the dyadic analysis when the number of active dyads crosses the registered threshold.

Deferred with explicit triggers are the digital-twin simulation program with uncertainty quantification, cross-cultural invariance testing, and the beamformer analysis path — each named in the backlog with the resource condition that would activate it. External validity, like everything else in v3.1, is a scheduled obligation with a trigger, not an aspiration with a footnote.

12. Disposition Ledger

This chapter is the revision's accountability record: one row per review point, 240 rows in total, each carrying a verdict and a pointer to the change that implements it. The ledger is organized in three parts — a summary of verdicts by section, the point-by-point disposition of the 144 defect findings, and the point-by-point disposition of the 96 enhancement recommendations — followed by the backlog of deferred items with their written trigger conditions. Resolution strings reference chapters, sections, equations, tables, and figures of this document, so any reader can walk from a reviewer's complaint to the exact normative text that answers it. Verdicts use the taxonomy of Table 1: ADOPT for full normative incorporation, PART for adoption with a documented residual constraint, CLAR for resolution by reclassification, and DEFER for backlog items whose activation conditions are recorded below. No point was rejected; where the review asked for something outside the framework's DIY, non-clinical constraints, the disposition says so explicitly rather than by omission.

Section	ADOPT	PART	CLAR	DEFER	N
Hardware (F-1..18)	14	4	0	0	18
Signal processing (F-19..42)	22	1	0	1	24
Ladder / NFB (F-43..60)	17	1	0	0	18
Channel/RV (F-61..84)	21	3	0	0	24
Gates / statistics (F-85..102)	17	1	0	0	18
Safety / ethics (F-103..120)	16	2	0	0	18
Conceptual (F-121..144)	15	2	6	1	24
Findings total	**122**	**14**	**6**	**2**	**144**

Table 9: Disposition summary — 144 defect findings.

Section	ADOPT	PART	CLAR	DEFER	N
Hardware (E-1..12)	9	3	0	0	12
Signal processing (E-13..28)	14	1	0	1	16
Protocol design (E-29..44)	15	1	0	0	16
Validation (E-45..60)	14	2	0	0	16
Safety / infrastructure (E-61..76)	12	4	0	0	16
Conceptual (E-77..96)	14	4	0	2	20
Enhancements total	**78**	**15**	**0**	**3**	**96**

Table 10: Disposition summary — 96 enhancement recommendations.

12.1 Ledger of Defect Findings (F-1 to F-144)

Each row records the reviewer's finding by its identifier, the verdict assigned in this revision, and the disposition — including the chapter, section, equation, table, or figure of v3.1 where the resolution lives. Readers checking the framework's work should sample against the source reports: identifiers preserve the reports' own numbering and section groupings, so F-37 is the thirty-seventh finding of the defects report and E-91 the ninety-first recommendation of the enhancements report.

ID	Verdict	Disposition and implementing change
F-1	**PART**	8-ch reclassified as screening tier; source claims restricted to 32-ch validation tier (Ch. 4.1)
F-2	**ADOPT**	Active/hybrid electrodes default; saline-gel upgrade path with documented SNR gains (Ch. 4.2)
F-3	**PART**	Gamma demoted to exploratory; operative gamma requires residual-EMG gate below 0.4 (Ch. 5.6)
F-4	**ADOPT**	Linked-mastoid for acquisition only; three-reference sensitivity battery mandatory (Ch. 4.5)
F-5	**PART**	250 Hz retained at screening tier; gamma-operative sessions require 500 Hz (Ch. 4.3)
F-6	**ADOPT**	Common-mode rejection test in bench QC; shielded cabling in build spec (Ch. 4.6)
F-7	**ADOPT**	Impedance threshold tightened to 25 kOhm; above-threshold channels excluded (Eq. 4.1)
F-8	**ADOPT**	Continuous impedance telemetry; per-block quality reports archived (Ch. 4.2)
F-9	**ADOPT**	Pressure-relief pads, torque-checked fit procedure, comfort self-reports (Ch. 4.7)
F-10	**ADOPT**	Active driver board + shielded ribbon default; mains residual measured (Ch. 4.2, 4.6)
F-11	**ADOPT**	Daisy tier restricted to sub-40 Hz features; gamma illegal on Daisy (Eq. 4.2)
F-12	**ADOPT**	Bipolar EOG + EMG aux mandatory; optional ECG; respiration belt at validation tier (Ch. 4.4)
F-13	**ADOPT**	USB preferred for trials; per-session drop/jitter budgets 0.5% / 1 ms (Ch. 4.3)
F-14	**ADOPT**	90-minute cap, wick liners, scalp temperature/humidity logging (Ch. 4.7)
F-15	**ADOPT**	Cost ladder includes consumables, calibration source, spares (Table 6)
F-16	**ADOPT**	Signal-generator calibration before first use and quarterly, hashed (Ch. 4.6)
F-17	**PART**	T7/T8 frame-limited, documented; auditory features out of scope (Table 5)
F-18	**ADOPT**	Version-controlled BOM, firmware pins, mechanical drawings (Ch. 4.8)
F-19	**ADOPT**	Inverse redefined as regularized forward fit; no source claims at 8 ch (Ch. 2.3)
F-20	**ADOPT**	P-hat re-anchored to SMR/beta; gamma/alpha ratio exploratory (Table 4)
F-21	**ADOPT**	MMN half-weighting removed; deviance battery with effect-size priors (Ch. 3.3)

ID	Verdict	Disposition and implementing change
F-22	**ADOPT**	Pair minimum expanded; ImCoh required for inter-network claims (Ch. 5.4)
F-23	**ADOPT**	Delay-discounting task fully specified end-to-end (Ch. 3.4)
F-24	**ADOPT**	Multi-state baseline + robust median/MAD normalization (Ch. 5.3)
F-25	**ADOPT**	Multitaper standard with 0.5 Hz resolution floor (Eq. 5.1)
F-26	**ADOPT**	ICA advisory only; ASR + aux-channel regression primary (Ch. 5.1)
F-27	**ADOPT**	PLV bias correction + 40-trial minimum + SNR gates (Eq. 5.4)
F-28	**ADOPT**	Surface Laplacian (validation) / nearest-neighbor (screening) added (Ch. 5.5)
F-29	**ADOPT**	Residual EMG index with gamma gate below 0.4 (Eq. 5.5)
F-30	**ADOPT**	PAC modulation index formalized with surrogate nulls (Eq. 5.6)
F-31	**ADOPT**	CI method fully specified: block bootstrap B=2000, pinned seeds (Ch. 8.2)
F-32	**ADOPT**	Version-pinned signed container with unit tests mandated (Ch. 5.8)
F-33	**PART**	Phase-randomization + shared-environment null both required (Ch. 5.4)
F-34	**ADOPT**	Family-explicit hierarchical FWER policy per gate (Ch. 8.3)
F-35	**ADOPT**	Delta-method error propagation into all estimator CIs (Eq. 2.6)
F-36	**ADOPT**	Online ASR-lite + excursion gating for feedback loop (Ch. 6.3)
F-37	**ADOPT**	IAF-individualized bands recomputed per session (Eq. 3.1)
F-38	**ADOPT**	Drift monitor with segment-wise renormalization (Ch. 5.3)
F-39	**ADOPT**	ImCoh / phase-slope index replace magnitude-squared coherence (Eq. 5.3)
F-40	**DEFER**	Beamformer path activates at funded 32-ch validation tier (Ch. 11.5)
F-41	**ADOPT**	T-hat gate requires two-task-block convergent evidence with CIs (Ch. 3.4)
F-42	**ADOPT**	Alpha-gamma interaction formalized as PAC or removed (Ch. 5.7)
F-43	**ADOPT**	Yoked sham mandatory from first L1 cohort (Ch. 6.2)
F-44	**ADOPT**	Boundary-state language removed; L2 repositioned as relaxation protocol (Ch. 6.4)
F-45	**ADOPT**	Thresholds tied to pilot CIs; adaptive per-operator calibration (Ch. 6.5)
F-46	**ADOPT**	Transfer test block required before level promotion (Ch. 6.5)
F-47	**ADOPT**	Sham control moved from EP-2 to L1 cohorts (Ch. 6.2)
F-48	**PART**	Load integral HRV-anchored where ECG present; self-report otherwise (Eq. 6.4)
F-49	**ADOPT**	Standardized recovery checklist + validated scales (Ch. 6.6)
F-50	**ADOPT**	CUSUM automated stop rule wired into feedback loop (Eq. 6.1)
F-51	**ADOPT**	Respiration/IMU covariates + partialling in dyadic coupling (Ch. 6.7)
F-52	**ADOPT**	300 ms loop latency budget + contingency verification (Ch. 6.3)
F-53	**ADOPT**	Lucidity rating replaced by standardized post-session report (Ch. 6.6)
F-54	**ADOPT**	Full blinding SOP: sealed allocation, blind experimenters, decoys (Ch. 7.4)
F-55	**ADOPT**	Exponential learning curve + dose-response models per level (Eq. 6.3)
F-56	**ADOPT**	Expectancy battery + yoked sham control practice effects (Ch. 6.2)
F-57	**ADOPT**	Adaptive thresholds with bounded floors and ceilings (Eq. 6.2)
F-58	**ADOPT**	DES-II screening, paired-operator rule, stop criteria (Ch. 9.2)
F-59	**ADOPT**	Learning-rate parameter with CI per operator (Eq. 6.3)
F-60	**ADOPT**	Artifact-gated scoring; movement strategies invalidate epochs (Ch. 6.3)
F-61	**ADOPT**	Priors reset conservative; expected-null retained; budget capped (Ch. 7.1)
F-62	**PART**	Automated 12-feature scoring primary; human judges consensus secondary (Ch. 7.3)
F-63	**ADOPT**	Certified hardware RNG, public seed commitment, per-session audits (Ch. 7.2)
F-64	**ADOPT**	Quantified isolation envelope + calibrated leakage measurement (Ch. 7.5)
F-65	**ADOPT**	ImCoh + environmental covariates + both surrogate nulls (Ch. 5.4, 6.7)
F-66	**ADOPT**	Trial layer resourced as capped negative-control program (Ch. 7.1)

ID	Verdict	Disposition and implementing change
F-67	**ADOPT**	Corrected-variance / block-bootstrap aggregation replaces Stouffer (Eq. 8.2)
F-68	**ADOPT**	Public audit log + external timestamp; deviation voids trial (Ch. 8.4)
F-69	**ADOPT**	Six external blind judges; conflicts declared; agreement stats (Ch. 7.3)
F-70	**ADOPT**	Semantic-distance floor + stratified randomization in pool design (Ch. 7.2)
F-71	**ADOPT**	Sketches hashed before feedback; scoring on sealed artifacts (Ch. 7.4)
F-72	**ADOPT**	12-feature automated vector pre-registered for free responses (Ch. 7.3)
F-73	**ADOPT**	Quantified isolation spec verified with instruments (Table 7)
F-74	**ADOPT**	Sync markers with 1 ms RMS drift budget (Ch. 7.6)
F-75	**ADOPT**	Positive-control condition with calibrated micro-expression dose (Ch. 7.7)
F-76	**ADOPT**	Camera + audio leakage measurement in positive-control arm (Ch. 7.7)
F-77	**ADOPT**	Entry bands reclassified as hypotheses; per-operator calibration (Ch. 7.8)
F-78	**PART**	20-min cap retained as safety default; documented titration protocol (Ch. 9.4)
F-79	**ADOPT**	Composite family-wise policy across FR/FC endpoints (Ch. 8.3)
F-80	**PART**	Registered reports mandated; community incentive policy out of scope (Ch. 8.5)
F-81	**ADOPT**	Retirement scoped to configuration + estimator version (Ch. 8.6)
F-82	**ADOPT**	Counterbalanced order; fatigue covariates pre-registered (Ch. 8.3)
F-83	**ADOPT**	Witness pool expanded to six; pairwise agreement reported (Ch. 7.3)
F-84	**ADOPT**	Power analysis: N=100 gives ~50% power at p1=0.33; registration raised to N=200 (Eq. 7.1)
F-85	**ADOPT**	ICC gate lowered to 0.4 with CI; 0.6 for promotable claims (Table 8)
F-86	**ADOPT**	8-week drift checkpoint added to reliability program (Table 8)
F-87	**ADOPT**	d 0.4 retained on multiplicity-protected primary (Table 8)
F-88	**ADOPT**	Yoked sham fully specified: contingency stripped, timing matched (Ch. 6.2)
F-89	**ADOPT**	EP-3 recalibrated to published hyperscanning effect ranges (Table 8)
F-90	**ADOPT**	Respiration/IMU partialling + inter-dyad control arm (Ch. 6.7)
F-91	**ADOPT**	Dual gate: z trigger + effect-CI / equivalence requirement (Table 8)
F-92	**ADOPT**	Retired flags propagate to dashboard; refusal infrastructure (Ch. 8.6)
F-93	**ADOPT**	Single registered interim look with alpha-spending (Ch. 8.4)
F-94	**ADOPT**	RFC-3161 external timestamps + OSF registration (Ch. 8.4)
F-95	**ADOPT**	External timestamp anchoring of audit hash chain (Ch. 8.4)
F-96	**ADOPT**	EP-5 multi-lab replication gate added (Table 8)
F-97	**ADOPT**	Exploration/confirmation firewall with versioned data splits (Ch. 8.5)
F-98	**ADOPT**	Bayesian sequential path with pre-specified priors (Ch. 8.2)
F-99	**ADOPT**	CI reporting mandatory in all results tables (Ch. 8.2)
F-100	**ADOPT**	Analysis-code freeze + variant manifest in container (Ch. 5.8, 8.4)
F-101	**ADOPT**	Parameter-recovery simulations mandated before fitting (Ch. 5.8)
F-102	**PART**	Open release required for external claims; consent-tiered internally (Ch. 8.5, 9.5)
F-103	**ADOPT**	Self-screen explicitly non-clinical; referral pathways documented (Ch. 9.1)
F-104	**ADOPT**	DES-II screening, paired-operator, stop criteria for L2 (Ch. 9.2)
F-105	**ADOPT**	Weekly cumulative load cap added (Eq. 6.4)
F-106	**PART**	Automated stop rules added; real-time clinical monitoring out of scope (Ch. 6.3, 9.1)
F-107	**ADOPT**	Young-adult protocol 18-21 with guardian consultation (Ch. 9.1)
F-108	**ADOPT**	Per-dataset granular revocable consent tiers (Ch. 9.5)
F-109	**ADOPT**	Ethics review required for external claims; self-assessment otherwise (Ch. 9.1)
F-110	**ADOPT**	Claims firewall + terminology policy across artifacts (Ch. 9.7, 10.3)
F-111	**ADOPT**	Stimulant window extended to 72 h; substance log as covariate (Ch. 9.3)

ID	Verdict	Disposition and implementing change
F-112	**ADOPT**	Sleep hours + validated mood scale logged per session (Ch. 9.3)
F-113	**PART**	Structured AE instrument adopted; passive monitoring remains manual (Ch. 9.6)
F-114	**ADOPT**	Clinical review precedes permanent retirement (Ch. 9.6)
F-115	**ADOPT**	Checksum chain + failover recorder + corruption quarantine (Ch. 9.6)
F-116	**ADOPT**	Uncertainty-first display; point estimates demoted (Ch. 10.2)
F-117	**ADOPT**	k-anonymity floor of 8 + salted release-scoped IDs (Ch. 9.5)
F-118	**ADOPT**	Retention schedule + 90-day deletion workflow (Ch. 9.5)
F-119	**ADOPT**	License headers + errata-register enforcement (Ch. 9.7)
F-120	**ADOPT**	Inherited layers demoted to interpretive status (Ch. 11.2)
F-121	**CLAR**	Axes are constructs; correlational status stated per estimator (Ch. 11.1)
F-122	**CLAR**	Hyper-correlation fork formalized: HC-S vs HC-P (Ch. 11.3)
F-123	**CLAR**	Reception posture relabeled operator-selected posture window (Ch. 11.1)
F-124	**ADOPT**	Comorbidity distance marked interpretive; no operative use (Ch. 11.2)
F-125	**CLAR**	Polytope/UHG moved to lineage appendix; non-load-bearing (Ch. 11.2)
F-126	**ADOPT**	xi(t) operationalized as residual variance with 0.35 gate (Eq. 2.7)
F-127	**ADOPT**	Explicit forward model specified (Eqs. 2.2-2.3)
F-128	**ADOPT**	Concrete solver + uniqueness condition (Eqs. 2.4-2.5)
F-129	**ADOPT**	Hierarchical self+cohort reference replaces (0,0,0) (Eq. 2.8)
F-130	**CLAR**	Prior tension documented; capped budget + written expectation (Ch. 11.4)
F-131	**ADOPT**	Predictive-coding mapping specified (Eq. 2.9)
F-132	**DEFER**	Mesh/game-theoretic extensions out of EEG scope; backlog trigger (Ch. 11.2)
F-133	**ADOPT**	All gates conditional on propagated CIs (Ch. 8.1-8.2)
F-134	**ADOPT**	Reference-swap + montage-subset sensitivity battery (Ch. 3.5, 4.5)
F-135	**ADOPT**	Weights reclassified as priors with credible ranges (Fig. 2)
F-136	**ADOPT**	Interactions formalized (PAC) or removed (Ch. 5.7)
F-137	**ADOPT**	10-20 capability comparison table added (Table 5)
F-138	**PART**	EP-5/EP-6 added; full sub-claim coverage on roadmap (Ch. 8.1, 11.5)
F-139	**ADOPT**	Red-team + adversarial collaboration institutionalized (Ch. 8.5)
F-140	**CLAR**	Charter revised; inherited layers non-operative (Ch. 1.1, 11.2)
F-141	**ADOPT**	Channel-capacity bound + data-processing inequality stated (Eq. 7.1)
F-142	**ADOPT**	HCI weights become fit parameters with sensitivity analysis (Ch. 5.7)
F-143	**ADOPT**	Masked-mode trials + debrief instrument (Ch. 10.3)
F-144	**PART**	External-validity roadmap added with triggered deferrals (Ch. 11.5)

Table 11: Point-by-point disposition of the 144 defect findings against EPF v3.0. ADOPT = normative change in v3.1; PART = adopted with documented constraint; CLAR = resolved by reclassification; DEFER = backlog with trigger condition.

12.2 Ledger of Enhancement Recommendations (E-1 to E-96)

The enhancement ledger follows the same grammar. Where an enhancement was adopted only partially — usually because it requires hardware the framework does not yet control, or because it applies only at the validation tier — the constraint is named in the resolution text so that the partial adoption is visible rather than implied. Deferred enhancements appear again in the backlog table below with their activation conditions.

ID	Verdict	Disposition and implementing change
E-1	**ADOPT**	Active/hybrid electrodes made the default build (Ch. 4.2)
E-2	**ADOPT**	Bipolar EMG/EOG aux channels mandatory at both tiers (Ch. 4.4)
E-3	**ADOPT**	Real-time impedance telemetry + adaptive gain enabled (Ch. 4.2)
E-4	**PART**	32-ch validation tier fully specified; funding-conditional (Ch. 4.1)
E-5	**ADOPT**	IMU motion co-registration at both tiers (Ch. 4.4)
E-6	**ADOPT**	Shielded-enclosure standards quantified in isolation spec (Table 7)
E-7	**PART**	USB-preferred policy; BLE permitted with measured drop/jitter budgets (Ch. 4.3)
E-8	**PART**	Open mechanical designs + fit-pressure protocol; sensor arrays deferred (Ch. 4.7)
E-9	**ADOPT**	Scalp temperature/humidity logging added (Ch. 4.6)
E-10	**ADOPT**	Pre-session calibration against signal generator (Ch. 4.6)
E-11	**ADOPT**	Saline/gel upgrade path with documented SNR deltas (Ch. 4.2)
E-12	**ADOPT**	Full BOM, firmware pins, drawings version-controlled (Ch. 4.8)
E-13	**ADOPT**	ImCoh / phase-slope index replace magnitude-squared coherence (Eq. 5.3)
E-14	**ADOPT**	IAF individualization of all band definitions (Eq. 3.1)
E-15	**PART**	Min-norm source estimates permitted at 32-ch tier only (Ch. 2.3)
E-16	**ADOPT**	Multitaper + wavelet estimators standardized (Eq. 5.1)
E-17	**ADOPT**	Delta-method error propagation into all CIs (Eq. 2.6)
E-18	**ADOPT**	Online ASR-lite for the real-time feedback loop (Ch. 6.3)
E-19	**DEFER**	Gamma validation against reference-grade systems; backlog trigger (Ch. 11.5)
E-20	**ADOPT**	PAC with surrogate controls operationalized (Eq. 5.6)
E-21	**ADOPT**	Volume conduction handled by ImCoh + reference policy (Ch. 5.4, 4.5)
E-22	**ADOPT**	Containerized, unit-tested pipeline with recovery tests (Ch. 5.8)
E-23	**ADOPT**	Hierarchical Bayesian estimators with shrinkage (Eq. 3.2)
E-24	**ADOPT**	Parameter-recovery simulations mandated pre-fit (Ch. 5.8)
E-25	**ADOPT**	Robust median/MAD normalization replaces mean/SD (Eq. 5.2)
E-26	**ADOPT**	Explicit forward-model approximation (Eqs. 2.2-2.3)
E-27	**ADOPT**	BIDS-EEG provenance logging (Ch. 5.8)
E-28	**ADOPT**	Automated residual-EMG quantification (Eq. 5.5)
E-29	**ADOPT**	Yoked sham arms at all neurofeedback levels (Ch. 6.2)
E-30	**ADOPT**	Pre-registered frozen analysis code (Ch. 8.4)
E-31	**ADOPT**	Positive-control subtle-cue condition (Ch. 7.7)
E-32	**ADOPT**	Calibrated leakage sensors, quarterly verification (Ch. 7.5)
E-33	**ADOPT**	Automated free-response scoring primary (Ch. 7.3)
E-34	**ADOPT**	Registered-report path for EP publications (Ch. 8.5)
E-35	**ADOPT**	EP-5 multi-lab replication gate (Table 8)
E-36	**ADOPT**	Bayesian sequential analysis with priors (Ch. 8.2)
E-37	**ADOPT**	Sleep/caffeine/mood covariates logged (Ch. 9.3)
E-38	**ADOPT**	Expectancy + demand questionnaires in battery (Ch. 6.2)
E-39	**ADOPT**	Exact acoustic/EM/olfactory criteria (Table 7)
E-40	**ADOPT**	Certified RNG + public seed + auditable selection (Ch. 7.2)
E-41	**ADOPT**	Independent AE monitor for multi-operator programs (Ch. 9.6)
E-42	**ADOPT**	Learning-curve + dose-response models (Eq. 6.3)
E-43	**ADOPT**	Null-model simulations for PLV/HCI (Ch. 5.4)
E-44	**PART**	Consent-tiered open release of data and code (Ch. 8.5, 9.5)

ID	Verdict	Disposition and implementing change
E-45	**ADOPT**	EP-1 thresholds with explicit CI reporting (Table 8)
E-46	**ADOPT**	Power analysis for small effects (Eq. 7.1)
E-47	**ADOPT**	Site/cohort random effects in cohort models (Ch. 8.2)
E-48	**ADOPT**	Effect-size CIs mandatory in all tables (Ch. 8.2)
E-49	**ADOPT**	TOST equivalence testing for null claims (Eq. 8.1)
E-50	**ADOPT**	Sensitivity analyses on constants and weights (Ch. 3.5)
E-51	**ADOPT**	Measurement-invariance testing across subgroups (Ch. 8.2)
E-52	**PART**	WAIC/LOO when competing models are pre-registered (Ch. 8.2)
E-53	**ADOPT**	OSF registration + external timestamp (Ch. 8.4)
E-54	**ADOPT**	Adversarial collaboration before claim-flipping gates (Ch. 8.5)
E-55	**ADOPT**	CONSORT-style flow diagrams for arms (Ch. 8.5)
E-56	**PART**	HRV required now; actigraphy tier-dependent; eye later (Ch. 6.6, 9.3)
E-57	**ADOPT**	Biomarker-behavior correlation tests in validation (Ch. 3.5)
E-58	**ADOPT**	Medication/substance covariates + extended windows (Ch. 9.3)
E-59	**ADOPT**	Pre-specified responder definitions (Ch. 6.5)
E-60	**ADOPT**	Living prediction registry (Ch. 8.6)
E-61	**ADOPT**	Ethics review for external claims (Ch. 9.1)
E-62	**ADOPT**	CUSUM automated stop rules (Eq. 6.1)
E-63	**ADOPT**	Validated dissociation/mood scales in recovery (Ch. 9.2)
E-64	**ADOPT**	Documented clinical referral pathways (Ch. 9.1)
E-65	**ADOPT**	Granular revocable consent tiers (Ch. 9.5)
E-66	**PART**	k-anonymity + salting now; DP for shared aggregates (Ch. 9.5)
E-67	**PART**	DSMB role for multi-operator programs (Ch. 9.6)
E-68	**ADOPT**	AE rates published annually in errata register (Ch. 9.6)
E-69	**ADOPT**	Equipment-failure/data-integrity protocol (Ch. 9.6)
E-70	**ADOPT**	Research dashboard separated from public claims (Ch. 10.3)
E-71	**PART**	90-day follow-up added; long-term program deferred (Ch. 9.6)
E-72	**ADOPT**	Commercial psychic framing prohibited in falsifiable mode (Ch. 9.7)
E-73	**ADOPT**	Signed version-pinned analysis containers (Ch. 5.8)
E-74	**ADOPT**	External timestamp anchoring RFC-3161 (Ch. 8.4)
E-75	**ADOPT**	Retraction/revision policy + errata register (Ch. 8.6, 1.3)
E-76	**PART**	Advisory pool defined; activation funding-conditional (Ch. 8.5)
E-77	**ADOPT**	Predictive-coding mapping (Eq. 2.9)
E-78	**ADOPT**	Information-theoretic bounds on transfer (Eq. 7.1)
E-79	**ADOPT**	Forward model operationalized (Ch. 2.2)
E-80	**PART**	Computational-psychiatry battery mapping at validation cohort (Ch. 11.5)
E-81	**PART**	HRV fusion operative now; eye-tracking deferred (Ch. 6.6, 11.5)
E-82	**DEFER**	Digital twin with uncertainty quantification; backlog (Ch. 11.5)
E-83	**PART**	Dyad-pool graph metrics at registered dyad threshold (Ch. 6.7, 11.5)
E-84	**ADOPT**	Positive controls as primary endpoints (Ch. 7.7)
E-85	**ADOPT**	Negative results published with equal prominence (Ch. 8.5)
E-86	**ADOPT**	BIDS-EEG / open-neuro alignment (Ch. 5.8)
E-87	**DEFER**	Cross-cultural/demographic invariance testing; backlog (Ch. 11.5)
E-88	**ADOPT**	MNE/EEGLAB validated wrappers with round-trip tests (Ch. 5.8)
E-89	**ADOPT**	Equation mutation and versioning rules (Ch. 1.3)

ID	Verdict	Disposition and implementing change
E-90	**ADOPT**	Synthetic benchmark datasets in container suite (Ch. 5.8)
E-91	**ADOPT**	EP-6 red-team exercises on isolation/scoring (Table 8)
E-92	**ADOPT**	Quantitative leakage models per channel (Ch. 7.5)
E-93	**ADOPT**	Hyperscanning best-practice alignment (Ch. 6.7)
E-94	**PART**	Cost ladder + feasibility metrics per tier (Table 6)
E-95	**ADOPT**	Public errata/amendment register (Ch. 1.3)
E-96	**ADOPT**	Independent replication before operative status (Ch. 8.1)

Table 12: Point-by-point disposition of the 96 enhancement recommendations. Verdicts as in Table 11.

12.3 Backlog and Trigger Conditions

Five items were deferred rather than adopted, each because it requires a resource the framework does not currently control. Deferral is not rejection: each item carries a written trigger condition, and when the condition is met the item graduates to the next revision's adoption queue automatically, without needing a fresh review cycle. The backlog is part of the living registry (Section 8.6) and its state is auditable alongside the gates.

Item	Substance	Trigger condition
F-40 / E-19	Beamformer source analysis; validation of gamma estimates against reference-grade acquisition	32-channel validation tier funded and commissioned
F-132	Game-theoretic and mesh extensions linked to the EEG layer	At least one operative estimator at EP-5 replication status
E-82	Digital-twin simulation program with uncertainty quantification	Validation cohort running with archived parameter-recovery corpus
E-87	Cross-cultural and demographic invariance testing	Operator pool exceeds 40 across at least three regions

Table 13: Deferred items with written trigger conditions.

With the ledger closed, the revision returns to its standing obligations: the gates of Chapter 8 remain the only path by which a v3.1 estimate becomes a v3.1 claim, the errata register remains the only mechanism by which this document changes, and the expected-null clause remains the framework's honest forecast for its own most dramatic layer. Revision R1's promise is narrow and checkable — every review point answered in a traceable row, every answer implemented in named text — and the specification now asks of its next reviewers only what it asks of itself: that claims keep pace with evidence, and stop when the evidence does.